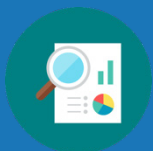


Unit 1 Overview: Number Sense and Operations With Positive Numbers

Unit Narrative



In this unit, students begin by exploring methods for finding the least common multiple (LCM) and greatest common factor (GCF) using the distributive property and other strategies. Students then build upon previous practice with decimal and fraction multiplication and division by connecting visual models to the standard algorithm and practicing a variety of strategies to improve arithmetic accuracy. This unit builds upon previously learned skills in 5th grade, including fraction and decimal operations, factors and multiples, and the distributive property. Later in this course and in the 7th grade accelerated course, students will use the skills from this unit to write equivalent expressions and equations and solve two-step equations.

The first lesson of Section 1 introduces students to the concept of the GCF using the distributive property. In the second lesson, students then explore the LCM. This section focuses on using a variety of strategies to find the GCF and LCM and uses both mathematical and real-world contexts to strengthen conceptual understanding.

Section 2 extends students' conceptual understanding from Section 1 to use the distributive property and the GCF of numbers to rewrite numeric expressions. Lesson 3 gives a general overview and scaffolded practice to rewrite both multiplicative and additive expressions, then provides additional practice specifically to rewrite sums using the distributive property. This section also incorporates models like the area model, pictured here, to visually represent equivalent expressions.

$$\begin{array}{|c|c|} \hline & + \\ \hline 27 & 18 \\ \hline \end{array}$$

Section 3 positions students to build upon previous practice with decimal and fraction operations to improve accuracy and fluency. The first two lessons focus on multiplying and dividing decimals, emphasizing the connections between the standard algorithm to visual representations and partial products. The last two lessons review the process for multiplying and dividing fractions and make connections between division and multiplication as students learn to divide fractions by multiplying by the reciprocal. This section extends previous learning by using fractions greater than one and decimals up to the thousandths place.

Unit At-A-Glance

Section 1: GCF and LCM	Benchmarks Addressed	Lesson 1	Finding the Greatest Common Factor
	MA.6.NSO.3.1	Lesson 2	Finding the Least Common Multiple
Section 2: Distributive Property	Benchmarks Addressed	Lesson 3	Rewriting Expressions Using the Distributive Property
	MA.6.NSO.3.2		
Section 3: Fluency with Operations	Benchmarks Addressed	Lesson 4	Multiplying Multi-Digit Numbers with Decimals to the Thousandths
	MA.6.NSO.2.1 MA.6.NSO.2.2	Lesson 5	Dividing Multi-Digit Numbers with Decimals to the Thousandths
		Lesson 6	Multiplying Positive Fractions
		Lesson 7	Dividing Positive Fractions



Differentiation

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Engagement <i>The “WHY” of learning</i>	- Prompt students to think about situations in which they would use multiplication and division of decimals in their everyday lives in Lessons 4 and 5. - Provide an opportunity for students to share their Wrap-up reflection in Lesson 6. Discuss actions and strategies that students took to get to “I feel confident and ready to go with...” for the concept they identified. By sharing their reflections, students both encourage others and reflect on their own growth mindset by using language that fosters community, perseverance, and resilience when working on challenging math problems or concepts.
Representation <i>The “WHAT” of learning</i>	- Utilize math manipulatives to represent real-world situations when finding GCF in Lesson 1. This physical and visual representation may support students in creating and seeing equal-sized groups using the GCF. - Use visual models like number lines or area models throughout the unit to connect algorithmic approaches to mathematical models. These models may be especially helpful for English Language Learners who may need to visualize a relationship between values without labels or vocabulary-rich real-world contexts yet.
Action & Expression <i>The “HOW” of learning</i>	- Consider creative opportunities for students to demonstrate their understanding of the differences between GCF and LCM. Venn Diagrams, Frayer Models, and other creative tools can be alternative methods for students to display their knowledge in a way that makes sense in their minds. - Consider trying ideas like stations, scavenger hunts, problem cards, or relay races to add variety and movement to student practice in Your Turn! sections in this unit.

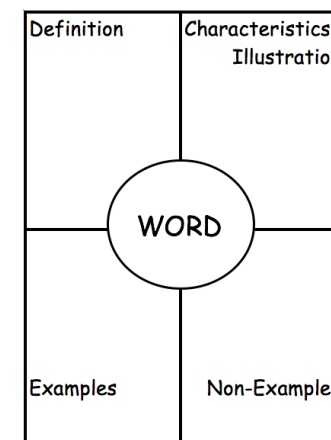
ELL Information

Students will be identifying and interpreting key features of graph of functions. This unit provides an opportunity to review some important vocabulary. Have students create a Frayer Model for the words in this topic:

- A Frayer Model (similar to the four-square graphic organizer) is a helpful graphic organizer because it allows students to see terms in multiple ways. For this section, students will be introduced to many words that set the foundation for functions.
- Provide the definition, but have students work in pairs to illustrate the word (when possible, or else list characteristics), list examples, and list non-examples.
- For examples and non-examples, encourage students to consider this a work in progress, and to add throughout the section as they are presented.
- Consider creating a Frayer Model for each term from the Glossary by Section lists (below).

All glossary terms are available in multiple languages, including English, Spanish, Haitian Creole, and American Sign Language, as support for ELL students. Consider pairing the following videos with strategies from “Additional Differentiated Support.”

- Greatest Common Factor
- Least Common Multiple
- Distributive Property
- Factors
- Multiples



Additional Differentiated Support

Scaffolding Resources	On-Ramp
Mark-Up Text can be used to help students approach more complicated and wordy problems, as well as to provide visual cues for additional feedback between students and teachers. Teacher will use a word problem where students will be able to identify which parts would need to be represented in numbers, and how to write those numbers in an expression or equation. • Teacher can provide students with suggested symbols for marking the text, such as those described below, or students can determine their own symbols to use as they mark up and make sense of the text. ✓ - background knowledge, they know what to do ? - confusing information; is this needed to solve the problem? ! - marks where a variable is needed to represent a number + - extra info in word problem; not needed • Using a small piece of text, teacher will model for students how you would Mark Up before releasing them to do the same. • Students will read the text slowly and carefully, focusing on one section at a time and marking text based on their key. Then they will turn to a partner to discuss each section and compare where they made marks and what information they determined in making them. • In closing, the teacher will select students to share their marks with the class at the end of the activity.	On-Ramp to 6th Grade Math provides video and practice tools for students to scaffold the following skills: • Adding Two-Digit and One-Digit Numbers • Addition Fluency • Subtraction Fluency • Prime and Composite Numbers • Distributive Property • Multiplying Decimals • Dividing Decimals • Multiply a Fraction by a Whole Number • Multiplying Fractions and Finding Area • Fractions in Real-World Multiplication Problems • Decimals in Expanded Form



Section Overview

Section 1: GCF and LCM (Lessons 1-2)

Benchmarks Addressed	Learning Targets	
MA.6.NSO.3.1 Given a mathematical or real-world context, find the greatest common factor and least common multiple of two whole numbers. <i>Clarification 1: Within this benchmark, expectations include finding greatest common factor within 1,000 and least common multiple with factors to 25.</i> <i>Clarification 2: Instruction includes finding the greatest common factor of the numerator and denominator of a fraction to simplify the fraction.</i>	Lesson 1	- I can find the greatest common factor (GCF) in mathematical and real-world contexts. - I can rewrite fractions in simplest form using the greatest common factor (GCF) of the numerator and denominator.
	Lesson 2	I can find the least common multiple (LCM) in mathematical and real-world contexts.

Section 2: Distributive Property (Lesson 3)

Benchmarks Addressed	Learning Targets	
MA.6.NSO.3.2 Rewrite the sum of two composite whole numbers that have a common factor, as a common factor multiplied by the sum of two whole numbers. <i>Clarification 1: Instruction includes using the distributive property to generate equivalent expressions.</i>	Lesson 3	- I can use the distributive property to generate equivalent numeric expressions. - I can rewrite the sum of two composite numbers with a common factor using distributive property.

Section 3: Fluency with Operations (Lessons 4-7)

Benchmarks Addressed	Learning Targets	
MA.6.NSO.2.1 Multiply and divide positive multi-digit numbers with decimals to the thousandths, including using a standard algorithm with procedural fluency. <i>Clarification 1: Multi-digit decimals are limited to no more than 5 total digits.</i> MA.6.NSO.2.2 Extend previous understanding of multiplication and division to compute products and quotients of positive fractions by positive fractions, including mixed numbers, with procedural fluency. <i>Clarification 1: Instruction focused on making connections between visual models, the relationship between multiplication and division, reciprocals and algorithms.</i>	Lesson 4	I can fluently multiply multi-digit numbers with decimals to the thousandths.
	Lesson 5	I can fluently divide multi-digit numbers with decimals to the thousandths.
	Lesson 6	I can fluently multiply positive fractions by positive fractions.
	Lesson 7	I can fluently divide positive fractions by positive fractions.



Vertical Alignment

Grade 6 Accelerated Unit 1
Number Sense and
Operations with Whole
Numbers



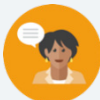
Grade 7 Accelerated Unit 14
Equivalent Algebraic
Expressions

Prior Course(s)	Course Level	Future Course
MA.4.AR.3.1 Determine the factor pair for a whole number from 0 to 144. Determine whether a whole number from 0 to 144 is prime, composite, or neither. MA.5.NSO.2.5 Multiply and divide a multi-digit number with decimals to the tenths place by one-tenth and one-hundredth with procedural reliability. MA.5.FR.2.2 Extend previous understanding of multiplication to multiply a fraction by a fraction, including mixed numbers and fractions greater than 1, with procedural reliability.	MA.6.NSO.3.1 Given a mathematical or real-world context, find the greatest common factor and least common multiple of two whole numbers. MA.6.NSO.3.2 Rewrite the sum of two composite whole numbers that have a common factor as a common factor multiplied by the sum of two whole numbers. MA.6.NSO.2.1 Multiply and divide positive multi-digit numbers with decimals to the thousandths place, including using a standard algorithm with procedural fluency. MA.6.NSO.2.2 Extend previous understanding of multiplication and division to compute products and quotients of positive fractions by positive fractions, including mixed numbers, with procedural fluency.	MA.8.AR.1.3 Rewrite the sum of two algebraic expressions having a common monomial factor as a common factor multiplied by the sum of two algebraic expressions.

1.1 Finding the Greatest Common Factor

Lesson Introduction

 Benchmark	MA.6.NSO.3.1 Given a mathematical or real-world context, find the greatest common factor and least common multiple of two whole numbers.		 Learning Targets	- I can find the greatest common factor (GCF) in mathematical and real-world contexts. - I can rewrite fractions in simplest form using the greatest common factor (GCF) of the numerator and denominator.
 Benchmark Coherence	Building On MA.4.AR.3.1 Determine factor pairs for a whole number from 0 to 144. Determine whether a whole number from 0 to 144 is prime, composite, or neither.		Working Toward MA.8.AR.1.3 Rewrite the sum of two algebraic expressions having a common monomial factor as a common factor multiplied by the sum of two algebraic expressions.	
 Essential Questions	- What is a greatest common factor? - What strategies can be used to find the greatest common factor of two whole numbers? - How can the greatest common factor be used to rewrite a fraction in simplest form?			
 Lesson Narrative	Students extend prior knowledge of finding factors of whole numbers to introduce how to find the greatest common factor (GCF) of two whole numbers. The lesson begins with an anchor task to introduce the concepts and definitions within a real-world context. Students then use the GCF to transition from informal “simplifying” to formally rewriting equivalent expressions, focusing on why the original fraction and the “simplest” form fraction are equivalent. Each new concept is followed by a brief Try It that students can do independently or with a partner.			
 Component Summary	1.1.1 Warm-Up (~5 min.)	Math Symbols Video (SE p. 3)	1.1.4 Guided Instruction (10 min.)	GCF in the context of numerator and denominator for reducing fractions (SE p. 6)
	1.1.2 Guided Instruction (10-15 min.)	Using a table to identify GCF and factor pairs (SE p. 4)	1.1.5 Try It (5 min.)	Simplifying and explaining equivalent expressions (SE p. 7)
	1.1.3 Try It (5-10 min.)	Finding GCF within a real-world context (SE p. 6)	1.1.6 Your Turn! (5-10 min.)	Simplifying fractions using GCF or common factors (SE p. 8)
	1.1.7 Wrap-Up (~5 min.)	Application problems, one skill-based and one within real-world context (SE p. 8)		
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> - Factor - Greatest common factor (GCF) - product <u>Additional Terms:</u> N/A		(Optional) Manipulatives for 1.1.2 Guided Instruction	1.1.1 Warm-Up contains a video to accompany the question prompts.

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may get stuck on finding factors for larger numbers. - Students may not recognize that they may not need to find all factors for both numbers to find GCF. - Students may not recognize that when dividing a numerator and denominator by the GCF, the only remaining common factor is 1.	- Consider making a word wall or anchor chart for vocabulary words introduced in this lesson, including <i>GCF</i> and <i>simplest form</i>, and to review any related previously learned vocabulary words. Add to this vocabulary chart after <i>multiples</i> are introduced later in Lesson 2, which can help clarify the difference between factors and multiples and minimize confusion. - Consider reviewing other strategies, including using organized lists or using divisibility rules to identify factor pairs.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Think-Pair-Share 1.1.2 and 1.1.4 Guided Instructions give students an opportunity to consider different real-world contexts in which the GCF is used. Consider giving students the opportunity to think of another situation in which the GCF is used to make equal sizes of groups throughout the lesson, then pair with a partner to share.	MA.K12.MTR.7.1: Real-World Contexts Students will find the GCF of two numbers within real-world contexts in the both the Guided Instruction and Try It portions of this lesson. Consider highlighting the connection of GCF to everyday experiences using a Think-Pair-Share. Other possible real-world contexts could include packaging, making teams, and grouping items.
 Differentiate Instruction	Consider finding the GCF of smaller numbers during Try It as an entry point for students who are still building fluency with recognizing factor pairs and common factors.	
	Consider using manipulatives to represent “snacks” and “letters” while completing the table from 1.1.2 Guided Instruction to help students visualize and physically create equal-sized groups using the GCF.	
If time permits, in the Wrap-Up consider having students create their own version of a real-world context problem that includes using the GCF to create equal-sized groups.		
Lesson Notes:		

1.1.1 Warm-Up: Where Do Math Symbols Come From?

Component Overview: Display the “Where Do Math Symbols Come From?” video for the entire class. After students watch this two-minute video, prompt the students to read and answer the questions in the Warm-Up independently or in partners.



1.1.1 Warm-Up: Where Do Math Symbols Come From?

Math is full of symbols that include lines, dots, arrows, English letters, Greek letters, and more. It can look and sometimes feel like its own language, and in many ways it is!

Robert Recorde invented the equals sign, $=$, because he was tired of writing “is equal to” over and over again.



- What is a word or phrase that you use a symbol to represent instead of writing it out each time?
Answers will vary. I use “&” to represent “and.” Instead of writing “percent,” I use the percent symbol (%).
- Why did early mathematicians decide to use symbols to express operations, phrases, and mathematical ideas?
Answers will vary. Mathematicians may have wanted to avoid using a lot of words, especially when sharing ideas with mathematicians who speak different languages. Symbols were used to organize mathematical ideas in a clear way.
- Why are math symbols still helpful to us today?
Answers will vary. Math symbols help us express mathematical ideas across language barriers. Symbols reduce the amount of writing that is needed.



After the Warm-Up, the teacher can have a few students share symbols they wrote for question 1 or have students share their answers to question 3 to build on the understanding of why symbols are useful in math. The discussions should be brief and not take time away from core instruction.



To increase engagement, consider having students record the math symbols while watching the video, categorizing them into symbols they know and symbols they do not know.

1.1.2 Guided Instruction: Greatest Common Factor

Component Overview: Students will be exploring an anchor task to find the GCF in the real-world context of creating care packages. Using a table to organize factor pairs, students will see common factors and be guided to identify the greatest number of care packages that is created using both snacks and letters. Guided Instruction positions the teacher to “think aloud” as students begin to formalize a conceptual understanding of GCFs. The teacher should identify and formalize the definitions of factor and GCF during this activity.



1.1.2 Guided Instruction: Greatest Common Factor

1. Marcus and Tonya are making care packages for soldiers overseas. Inside of each package, they will include popular snacks and handwritten letters. They have 16 snacks and 12 letters to put in the care packages. They want each care package to be equally sized, containing both snacks and letters, with nothing left over.
 - a. Complete the tables below to show the possible combinations when making each care package.

Distributing Snacks Equally	
Number of Care Packages	Number of Snacks per Package
1	16
2	8
4	4
8	2
16	1

Distributing Letters Equally	
Number of Care Packages	Number of Letters per Package
1	12
2	6
3	4
4	3
6	2
12	1

- b. What is the greatest number of care packages that can be created using **only** snacks?
16 care packages
 - c. What is the greatest number of care packages that can be created using **only** letters?
12 care packages
 - d. What is the greatest number of care packages that can be created using **both** snacks and letters?
4 care packages



Consider using manipulatives to represent the snacks and letters. Place 16 “snacks” and 12 “letters” on a display or table visible to all learners, then physically move the manipulatives to make equal-sized groups as learners complete the table. Emphasize that equal groups are required for the care packages to be complete.



Questions 1a and 1b inform students’ understanding of what it means by using ‘both’ in question 1d.

- How are the numbers in each column for the snacks table related to 16?
- How are the numbers in each column for the letters table related to 12?
- For question 1d: Can 2, 3, 4, 5, 6, 8, 12, or 16 care packages be created using both snacks and letters?

1.1.2 Guided Instruction: Greatest Common Factor (continued)



A **factor** is a number that divides into another number exactly and without leaving a remainder.

For example, the **factors** of 20 are 1, 2, 4, 5, 10, and 20.

The numbers (or expressions) that are being multiplied are called the factors. When numbers (or expressions) are multiplied, the answer is called a product.



factor $\times$ **factor** = **product**

- e. How were the factors of 12 and 16 used in the previous problem?
- The factors of 16 were used to determine the whole number of ☐ letters ☒ snacks that could be included, depending on the number of care packages.
 - The factors of 12 were used to determine the whole number of ☒ letters ☐ snacks that could be included, depending on the number of care packages.



The **greatest common factor**, or **GCF**, is the largest common factor of two or more numbers.



Think-Pair-Share

Consider discussing that making care packages is only one real-world context and probe students to think of another scenario in which the GCF is used to make equal-sized groups. Possible other contexts include making equal-sized groups, packaging, or teams. Emphasize conceptual connections between the real-world contexts requiring the creation of equal-sized groups.

1.1.2 Guided Instruction: Greatest Common Factor (continued)

An example of how to find the **greatest common factor (GCF)** is shown.

The factors of 15 are 1, 3, 5, and 15.

The factors of 20 are 1, 2, 4, 5, 10, and 20.

The GCF of 15 and 20 is 5.

- f. Complete the statement to explain what the GCF of 16 and 12 is from the previous problem.

The GCF of 16 and 12 is 4, which represents the greatest number of care packages that can be created using

- ☐ only snacks.
- ☐ only letters.
- ☒ both snacks and letters.

2. Find the greatest common factor of 6 and 18.

	Factors	GCF:
6	1,2,3,6	6
18	1,2,3,6,9,18	



How can you check that all the factor pairs for two numbers have been found?



For question 2, consider using an organized list (vertical or horizontal) to visually show all factor pairs. To engage visual learners in seeing factor pairs, list the factors from least to greatest and consider using a line to connect the factor pairs. Then circle the common factors between the two lists to help visually identify which is the greatest.

1.1.3 Try It: Greatest Common Factor

Component Overview: Students practice finding the GCF of pairs of numbers by using factor pairs. Question 2 introduces the real-world concept of creating equal-sized teams, extending students' understanding of finding and using GCF in everyday experiences.



1.1.3 Try It: Greatest Common Factor

- Find the greatest common factor of each pair of numbers.

a. 15 and 200

$$\text{GCF} = 5$$

b. 123 and 41

$$\text{GCF} = 41$$

c. 54 and 36

$$\text{GCF} = 18$$

d. 282 and 108

$$\text{GCF} = 6$$

- Penelope is organizing the teacher-student relay race. The teams must be divided evenly, with no participants left out. If there are 56 students registered and 21 teachers, what is the greatest number of teacher-student teams that will be able to participate in the relay race?

7 teacher-student teams



Consider possible arrangements in the classroom to allow students to work in partners for added peer support or facilitate student agency by allowing students to choose to complete two or three of the four total questions. For movement, post each of these to a corner of the classroom and have students walk around to each corner while finding the GCF.



For questions 1a-d, consider asking students to discuss whether they need to find all factors to find the GCF. For example, in question 1a when finding the GCF of 15 and 200, they do not need to list every factor of 200. They could list the factors of 15 (a much shorter list) and from there determine the greatest of those that is also a factor of 200.



For question 2, what do you notice about the real-world context that helps you know we are looking for the GCF of these values?

1.1.4 Guided Instruction: Writing Equivalent Fractions Using GCF

Component Overview: Students review the key vocabulary of equivalent fractions and are introduced to using the GCF to rewrite fractions in simplest form. Students focus on reasoning related to identifying equivalent fractions and explaining that they are equivalent because they have the same value. Consider using this sample as a guide for a “think aloud” as the teacher, or place students in pairs and have them use the self-guided prompts to explore the process of GCF through the lens of fractions.



1.1.4 Guided Instruction: Writing Equivalent Fractions using GCF

The GCF can also be used to rewrite fractions. When using the GCF to write an equivalent fraction, the result is often said to be in “simplest” form.



To **simplify a fraction**, generate an equivalent fraction by dividing the numerator and denominator by the **GCF** of the numerator and denominator.

1. In the fraction $\frac{25}{90}$, the numerator is 25 and the denominator is 90.
 - a. What is the greatest common factor of 25 and 90?
GCF = 5
 - b. Divide the numerator and denominator by the GCF. What is the resulting fraction?
 $\frac{5}{18}$
 - c. Complete the statement to describe the relationship between the fractions.

After dividing the numerator and denominator of $\frac{25}{90}$ by their GCF, the resulting fraction, $\frac{5}{18}$, ☐ is equivalent to $\frac{25}{90}$, rewritten in simplest form. ☐ is not



What would be the new GCF of the remaining values? What does that tell you about when you know an answer is in “simplest” form?



For students who might finish early or have extra time while other students are still formalizing the concept, consider having students create their own example and switch with a partner to simplify using the GCF.

1.1.5 Try It: Writing Equivalent Fractions using GCF

Component Overview: Question 1 gives students the opportunity to (individually or in partners) write equivalent fractions using the GCF of the numerator and denominator. Question 2 focuses on reasoning and explaining why two fractions are equivalent using the GCF as the simplifying factor.



1.1.5 Try It: Writing Equivalent Fractions using GCF

- Rewrite each fraction in simplest form using the GCF of the numerator and denominator.

a. $\frac{8}{28}$

GCF = 4

$\frac{2}{7}$

b. $\frac{45}{90}$

GCF = 45

$\frac{1}{2}$

c. $\frac{16}{34}$

GCF = 2

$\frac{8}{17}$

- Mikayla states that $\frac{900}{1000}$ is not the same value as $\frac{9}{10}$, because they are fractions of different sizes. Is she correct? Provide an explanation to support your answer.

Answers will vary. She is incorrect. They are equivalent fractions; $\frac{900}{1000}$ can be reduced to $\frac{9}{10}$ by dividing the numerator and denominator by 100.



For extension on question 1a and 1b, give students the opportunity to notice the difference between generating equivalent fractions with a common factor that is not the GCF and using the GCF. Emphasize that both approaches are valid but using a common factor may take multiple steps. For example, in 1b, simplifying with the common factors 9 and 5 requires two steps, whereas simplifying with the GCF 45 takes one step.



For question 2, consider reviewing divisibility rules for numbers ending in zero or multiple zeros. This strategy may help students who are having difficulty identifying common factors for large numbers. Prompting could include, "What do you notice about the ending of 900 and 1000? How can you tell that both numbers are divisible by 10? By 100? What is the greatest common factor: 10 or 100?"

1.1.6 Your Turn!

Component Overview: Students practice both learning targets in these problems. Question 1 focuses on finding the GCF of a pair of numbers, while question 2 uses the GCF to simplify a fraction. Question 3 provides another example of a real-world scenario for using the GCF to create equal-sized groups.



1.1.6 Your Turn!

1. Find the greatest common factor (GCF) of each pair of numbers.

a. 18 and 24	b. 45 and 60	c. 665 and 30
$GCF = 6$	$GCF = 15$	$GCF = 5$

2. Rewrite each fraction in simplest form using the GCF of the numerator and denominator.

a. $\frac{29}{87}$	b. $\frac{10}{15}$	c. $\frac{175}{210}$
$GCF = 29$	$GCF = 5$	$GCF = 35$
$\frac{1}{3}$	$\frac{2}{3}$	$\frac{5}{6}$

3. Mrs. Chaunuan is making gift bags filled with pencils and stickers for class prizes. Mrs. Chaunuan has 24 pencils and 36 stickers to use. Each bag will have the same number of each item, with no items left over. What is the greatest number of gift bags she can make?

12 gift bags



Consider engaging students in discussion on how to use multiple approaches to finding GCF, including organized lists, finding factor pairs using known multiplication facts, finding factor pairs for the smaller of the two values first to compare common factors, and/or revisiting divisibility rules to quickly determine if factors are shared. Questioning could include, “How can we find the GCF using a different strategy?” or “Which strategy do you prefer? Why?”



For any part of question 1: What are the factors of ___ and ___? Which is the greatest common factor? Do I need to find all factors of both numbers to find the GCF?



Consider asking students to extend their thinking by drawing a visual model or diagram of the equal-sized groups of the pencils and stickers used for each bag.

1.1.7 Wrap-Up: Ticket Out the Door

Component Overview: Consider using all or parts of this exercise as a formative assessment to gather data on student progress from this lesson. This Wrap-Up assesses students' ability to rewrite fractions in simplest form using the GCF of the numerator and denominator and apply finding the GCF to a real-world context.



1.1.7 Wrap-Up: Ticket Out the Door

1. Rewrite $\frac{52}{104}$ in simplest form.

GCF = 52

$$\frac{1}{2}$$

2. Diego is preparing brownies and cookies for a bake sale. He has 48 brownies and 64 cookies. Diego wants to make bake sale bags. Each bag will have the same amount of each item, with no items left over. What is the greatest number of bake sale bags he can make?

16 bake sale bags

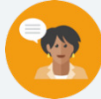


For visual learners, consider allowing students to represent their thinking for question 2 using a model or diagram. Students could draw or label a model of the brownies and cookies, then explain how they used the GCF to determine their answer.

1.2 Finding the Least Common Multiple

Lesson Introduction

 Benchmark	MA.6.NSO.3.1 Given a mathematical or real-world context, find the greatest common factor and least common multiple of two whole numbers.		 Learning Target	I can find the least common multiple (LCM) in mathematical and real-world contexts.
 Benchmark Coherence	Building On MA.4.AR.3.1 Determine factor pairs for a whole number from 0 to 144. Determine whether a whole number from 0 to 144 is prime, composite, or neither.		Working Toward MA.8.AR.1.3 Rewrite the sum of two algebraic expressions having a common monomial factor as a common factor multiplied by the sum of two algebraic expressions.	
 Essential Questions	- What is the least common multiple (LCM)? - What strategies can be used to determine the LCM of two whole numbers? - How does the LCM appear in real-world contexts? - What are the similarities and differences between finding the LCM and the GCF?			
 Lesson Narrative	This lesson introduces finding the least common multiple (LCM) of two whole numbers. The Warm-Up includes a video where students watch 3 sand timers, which then leads into the Guided Instruction around that scenario to build conceptual understanding of LCM in real-world contexts. Number line models are used to introduce multiples, common multiples, and the least common multiple. Then students are given an opportunity to practice finding the LCM to solve both real-world and mathematical problems.			
 Component Summary	1.2.1 Warm-Up (~5 min.)	Hourglass timing video and discussion (<i>SE p. 9</i>)	1.2.4 Collaboration (10 min.)	Open-ended task using digits 1-9 to create factors and LCM (can be completed on different day) (<i>SE p. 12</i>)
	1.2.2 Guided Instruction (10 min.)	Use number line models to connect the hourglass video to finding LCM (<i>SE p. 10</i>)	1.2.5 Your Turn! (10 min.)	Students find LCM of pairs of numbers and in real-world contexts (<i>SE p. 13</i>)
	1.2.3 Try It (10 min.)	Students find LCM of pairs of numbers and in real-world contexts (<i>SE p. 11</i>)	1.2.6 Wrap-Up (~5 min.)	Self-assessment rating of confidence in finding the GCF and LCM (<i>SE p. 14</i>)
 Resources	Glossary		Materials	
	<u>Key Terms:</u> - Multiples - Ellipses - Infinite - Least common multiple (LCM) <u>Additional Terms:</u> N/A		N/A	
			Additional Preparation	
			Warm-Up 1.2.1 contains a video to accompany the question prompts.	

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may confuse GCF and LCM. - Students may struggle to know when to start/stop their lists or with what to do if they do not see a common multiple. - Students may falsely generalize the LCM as always being the product of the two numbers.	- Consider drawing specific attention to the differences in finding the GCF in Lesson 1 to finding the LCM in this lesson. - Consider adding to the word wall or anchor chart from Lesson 1 by adding new vocabulary words introduced in this lesson, including LCM and multiples. Referencing an anchor chart can help clarify the difference between factors and multiples and minimize confusion.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Notice and Wonder Consider using the hourglass timing video in 1.2.1 Warm-Up and other real-world contexts throughout the lesson as an opportunity for students to notice and wonder about math in the real world in 1.2.2 Guided Instruction. Consider guiding students through “Notice and Wonder” routines as they ask questions and wonder about how the hourglasses work and connect to finding LCM in 1.2.4 Collaboration.	MA.K12.MTR.7.1: Real-World Contexts Students will find the LCM of two numbers within real-world contexts in the 1.2.2 Guided Instruction, 1.2.5 Your Turn!, and 1.2.3 Try It portions of this lesson. Consider highlighting the connection of LCM to everyday experiences. Possible other real-world contexts could include finding when multiples overlap related to time or distance.
 Differentiate Instruction	Consider using the problems within the 1.2.3 Try It section as an opportunity to find the LCM using a variety of strategies within student math journals or as an interactive notebook entry. This entry could be used to compare with similar work on the GCF in Lesson 1 and can be a reference for students throughout this unit to help clarify the difference between LCM and GCF.	
	Consider using the number line models from 1.2.2 Guided Instruction section as a support for visual learners in following sections.	
	During 1.2.5 Your Turn! consider having students create their own version of a real-world context problem that includes using the LCM to find when multiples overlap related to time or distance.	
Lesson Notes:		

1.2.1 Warm-Up: Hourglass Timing

Component Overview: Display the “Hourglass Timing” video for the entire class. The goal of this video is to get students to wonder about the situation presented: three different sand timers set to empty at different rates and wondering when all three timers will run out at the same time. After watching the video, first allow students to write down questions they have as they watch the video or view the images. Allow students to share with a partner, then as a class discuss how each timer is running at a different rate, which will result in a different number of flips during the 30-minute span of time for each timer.



1.2.1 Warm-Up: Hourglass Timing

The video shows three different hourglass sand timers. The timers are all started at the same time. As one of the times runs out, it is flipped over again to restart.



Several screenshots taken from the video are shown.



1. Write one question you have as you watch the video or review the images.

Answers will vary. What could we change about the timers, besides the amount of sand to make them empty quicker?

2. Compare questions with your partner.
3. Record your partner’s question in the table and have them initial that you recorded it correctly. If you had the same question, work with your partner to generate a different question to ask about the video. *You are the only person who should write in the first column of the table.*

My Partner’s Question From the Video	Initials
<i>Answers will vary. What could we change about the timers, besides the amount of sand to make them empty slower?</i>	



Notice and Wonder

After watching the video, consider implementing a Notice and Wonder routine using partner work in this activity. Encourage discourse by facilitating a conversation based on student questions and the video. Ideally, the conversation would lead toward a question that can frame the learning target on LCM, “When will all three timers have no sand in the top half at the same time?” Consider probing questions to guide students toward that framing question.

1.2.2 Guided Instruction: Least Common Multiple

Component Overview: Students explore the concept of finding the LCM with the three hourglass timers from the Warm-Up using a number line model. Be sure to read and emphasize definitions and examples of multiples and the LCM within the section progression. Consider guiding students through this section as a class to discuss different strategies, clarify any misconceptions, and have students write down definitions of multiples and the LCM in a math journal or notebook.



1.2.2 Guided Instruction: Least Common Multiple

1. Three timers were used in the video, a 2-, 3-, and 5-minute timer.
 - a. On each number line model, plot a point indicating the time at which each hourglass would need to be flipped over to restart the timer during a 30 minute period.
Answer shown in table.

Time	Points Where the Timer Needs to be Restarted
2-minute timer	
3-minute timer	
5-minute timer	

- b. What do you notice about the times plotted on the number line for each timer?
Answers will vary. I notice the overlap of some of the times. I notice that the 2-minute timer had the most points and needed to be flipped the most.

Multiples of positive whole numbers are the product of that number multiplied by a positive whole number.

For example, the **multiples** of 2 are 2, 4, 6, 8



The **ellipses** (...) at the end of the list of multiples means that the list continues beyond the last number listed. There are an **infinite** number of multiples for any given whole number.



The benchmark for this lesson includes finding the LCM of two numbers, so use questioning prompts to help students identify pairs of timers within the number line model. For example, when answering the question, “What are all the common multiples among the three timers?” include pairs of timers, as well as later looking at all three for added complexity.



Consider asking probing questions such as the following:

- What do the points on the line represent?
- How does this connect to the video we watched in the Warm-Up?
- How do these dots relate to one another among all three number lines?

1.2.2 Guided Instruction: Least Common Multiple (continued)

2. Use the number line models for each of the timers and the definition of multiples to answer the questions.

- a. What are all the common multiples among the three timers?

30 is the common multiple between all three timers.

- b. What do these common multiples represent in this scenario?

Answers will vary. When all the timers would be flipped at the same time. When the sand runs out for all the timers simultaneously.

- c. After how many minutes will the 2- and 5-minute timers run out of sand at the same time?

After 10, 20, and 30 minutes.

- d. After how many minutes will the 3- and 5-minute timers run out of sand at the same time?

After 15 and 30 minutes.



The **Least Common Multiple**, or **LCM**, is the lowest number that is a multiple of two or more given numbers.

3. Use the definition to complete the statement.

The least common multiple (LCM) of 2, 3, and 5 is 30.
In the context of the hourglass timers, this value represents when the timers will run out simultaneously.

4. Find the LCM of 8 and 6.

LCM = 24



Consider keeping the video on display for any of the parts in question 2. Revisiting the video may help visual learners answer the questions if they are able to re-watch important segments or see how the hourglasses flipping relates to the multiples on the number line model.



When modeling finding the LCM of 8 and 6, it is helpful to model using an organized list (vertical or horizontal) and thinking aloud about when to stop the list versus when to keep going. Also consider circling the LCM in both lists. This strategy coupled with the think aloud can serve as an entry point for both visual and auditory learners.

1.2.3 Try It: Least Common Multiple

Component Overview: Consider having students work in partners for support as they continue practicing using organized lists, number line models, or other strategies to determine the LCM of pairs of numbers and in real-world contexts. Another option is for students to record their work and answers in a math journal or interactive notebook page as a reference to look at in future lessons and to compare to finding the GCF in Lesson 1.



1.2.3 Try It: Least Common Multiple

1. Find the LCM of each pair of numbers.
 - a. 10 and 4
 $LCM = 20$
 - b. 5 and 13
 $LCM = 65$
 - c. 7 and 21
 $LCM = 21$
2. Every 4 weeks, Fort Caroline Middle has an early release day. Every 5 weeks, they host an assembly with a guest speaker for all students. After how many weeks will Fort Caroline Middle have an assembly and early release in the same week?
 20 weeks
3. A radio station is running two ongoing promotions. For the first promotion, every 30th caller wins a pair of movie tickets. For the second promotion, every 100th caller wins an annual pass to a theme park. Which caller number would be eligible for both prizes?
 300
4. Vincenzo and Marie both joined a local gym. Vincenzo plans to go to the gym every 2 days at 7:00 am when they open. Marie plans to go every 3 days at 7:00 am. If they both joined the gym on the same day, after how many days will they arrive at the gym on the same day?
 6 days



Help students analyze their responses in problem 1 and recognize that the LCM is not always the product of the two numbers. After students finish all three parts of question 1, ask them, “What do you notice about the LCMs of the first three pairs of numbers?” Be sure to point out that sometimes the LCM is the product of the two factors; sometimes one of the numbers itself is the LCM; and sometimes the LCM is larger than both numbers, but less than the product.



What are the multiples of ___ and ___? What are the common multiples of ___ and ___? Which is the least common multiple? How do I know when I have found enough for finding the LCM?

1.2.4 Collaboration: Least Common Multiple

Component Overview: Students work in partners to use the digits 1 through 9 one time each to make a statement about two factors and the LCM true. Consider setting a time limit for this portion of the lesson, as there are many correct answers and finding possibilities may take students more time than is available. If time is limited, this can also be used on a different day as a review or open-ended task.



1.2.4 Collaboration: Least Common Multiple

1. Place a digit in each blue box to make a true statement. Use only digits 0 to 9, at most one time each.

Factor A:

Factor B:

LCM:

Answers will vary. For example, 28 and 35: LCM = 140

2. What were some of the challenges you faced with this problem?
Answers will vary. Some of the LCMs were large numbers so finding them took a long time. It was challenging to not reuse digits.

3. What methods did you use to solve this problem?
Answers will vary. We wrote out all the multiples until we found a match.



Consider using probing questions while monitoring student collaboration for students who may need additional support:

- What do you notice about the number of boxes available?
- What do you notice about the LCM?
- What must be true for the factors and the LCM?
- What do you think is a good strategy to get started?
- Did you use a number more than once in any of the boxes?



This task can be complex because there are many correct answers, and the LCMs can be large values. To support learners, help students think aloud and verbalize what they notice before getting started, including recognizing that the factors are two digits and can include 0 as the first digit to represent a value with only ones. For example, 07 can be used to represent the factor of 7.

1.2.5 Your Turn!

Component Overview: Students individually apply strategies to finding the LCM for a pair of numbers and real-world problems. Have students complete this work independently as you circulate to check for student understanding.



1.2.5 Your Turn!

1. Find the LCM of each pair of numbers.

- a. 15 and 20 b. 18 and 24 c. 22 and 11
 $LCM = 60$ $LCM = 72$ $LCM = 22$

2. Jasmine and Teia are running a race. Jasmine runs at a rate of 4 minutes per lap and Teia runs at a rate of 6 minutes per lap. If they both start running at the same time, after how many minutes will they both be at the starting line at the same time again?

12 minutes

3. Lin's uncle is opening a bakery. On the bakery's grand opening day, he plans to give away prizes to the first 50 customers that enter the shop. Every 9th customer will get a free blueberry muffin. Every 12th customer will get a free slice of carrot cake. Which customer will get both free bakery items?

36 th customer



How do you know when you have found enough multiples to determine the LCM? How can you determine the LCM?



Consider encouraging students to use visual models to find the LCM for the word problems in this section. For example, students can use a number line model like in the Guided Instruction example. Students may also use organized lists and skip count to determine multiples, then circle the LCM as a visual strategy.



Consider extending student understanding of the LCM by asking students to think of other contexts, besides those in this section, where the LCM is used to solve a real-world problem, including when multiples overlap related to time, distance, or order.

1.2.6 Wrap-Up: Reflection

Component Overview: Use this self-evaluation as an opportunity for students to reflect on their level of mathematical confidence in understanding GCF and LCM from Lessons 1-2.



1.2.6 Wrap-Up: Reflection

1. Rate your level of mathematical confidence on the concepts listed by circling the statement that best describes your understanding.

Answers will vary.

- **Greatest Common Factor (GCF)**

I don't know what GCF is.	I need a little more practice.	I am okay at this.	I'm pretty good at this.	I could teach this.
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- **Least Common Multiple (LCM)**

I don't know what LCM is.	I need a little more practice.	I am okay at this.	I'm pretty good at this.	I could teach this.
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If time permits, consider asking students to explain the difference between the GCF and LCM with a partner before completing this self-evaluation. This practice of recalling the difference both supports conceptual understanding and provides an entry point for auditory learners in the self-evaluation.



For students who may finish early, consider having them provide examples of GFC and LCM to support their statements.

1.3 Rewriting Expressions Using the Distributive Property

Lesson Introduction

 Benchmark	MA.6.NSO.3.2 Rewrite the sum of two composite whole numbers having a common factor, as a common factor multiplied by the sum of two whole numbers.		 Learning Targets	- I can use the distributive property to generate equivalent numeric expressions. - I can rewrite the sum of two composite numbers with a common factor using the distributive property.
 Benchmark Coherence	Building On N/A		Working Toward MA.8.AR.1.3 Rewrite the sum of two algebraic expressions having a common monomial factor as a common factor multiplied by the sum of two algebraic expressions.	
 Essential Questions	- What are the similarities and differences between writing equivalent expressions using the area model and the distributive property? - How are common factors used to generate equivalent expressions for a sum of two composite numbers?			
 Lesson Narrative	Students will extend prior knowledge of the distributive property to area models for rewriting numeric expressions. The lesson includes rewriting multiplication expressions and addition expressions using the distributive property and factors to build stronger foundations of equivalent numeric expressions before delving deeper into rewriting sums of composite numbers with a common factor. Students collaborate and discuss strategies to find missing values to make equations, comparing equivalent numeric expressions and analyzing errors when rewriting expressions using the distributive property. The Guided Instruction and Collaboration in this lesson provide opportunities to encourage flexible math thinking by rewriting equivalent expressions in various ways, including with common factors and the GCF.			
 Component Summary	1.3.1 Warm-Up (~5 min.)	Students use clues to identify a “secret number” (SE p. 15)	1.3.4 Guided Instruction (10 min.)	Analyze thinking and errors when generating equivalent expressions (SE p. 17)
	1.3.2 Guided Instruction (10 min.)	Guiding prompts for applying the area model for representing distributive property (SE p. 15)	1.3.5 Collaboration (5-8 min.)	Work with a partner to find missing values in equations (SE p. 18)
	1.3.3 Guided Practice (5 min.)	Practice rewriting expressions using the distributive property and area model (SE p. 16)	1.3.6 Your Turn! (5-7 min.)	Generate multiple equivalent expressions using common factors and the GCF (SE p. 19)
	1.3.7 Wrap-Up (~5 min.)	Demonstrate understanding of learning targets by rewriting equivalent expressions using the distributive property (SE p. 19)		
 Resources	Glossary		Materials	
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> - Distributive property of multiplication over addition - Greatest common factor (GCF)		(Optional) Chart paper (Optional) Sticky notes (Optional) Markers (Optional) Visual anchor chart for distributive property as area model	
			Additional Preparation	
			If using a shared space for students to contribute ideas during 1.3.6 Your Turn!, prepare chart paper or a whiteboard space with the expressions from questions 1a-d. Gather sticky notes or markers for students to write generated equivalent expressions.	

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students who struggle with multiplication may struggle to identify factors and equivalent expressions. - Students may confuse factors and multiples. - Students with gaps in conceptual understanding of area may have difficulty using the area model to show the distributive property. - Students may struggle to differentiate between expressions and equations.	- Consider using material and video from Lesson 1 to work on common factors. - While Guided Instruction and Your Turn include prompts to rewrite expressions using the GCF, not every expression needs to be rewritten using the GCF. Encourage flexible thinking to rewrite expressions using both the GCF and other common factors. - Throughout the lesson, math vocabulary is used to identify parts of an expression and answers. Consider reviewing the vocabulary words <i>sum</i>, <i>product</i>, <i>addend</i>, and <i>factor</i> before or during instruction. - If time is an issue, consider assigning 1.3.6 Your Turn! for homework.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Clarify, Critique, and Correct 1.3.4 Guided Instruction provides an opportunity for students to develop math reasoning by analyzing errors when generating equivalent expressions. Consider prompting students to identify incorrect use of common factors or the distributive property, then correcting errors.	MA.K12.MTR.4.1: Discussion and Reflection 1.3.4 Guided Instruction positions students to analyze the math thinking of others and lets students recognize errors to suggest how to correctly rewrite an equivalent expression. Consider using these opportunities to engage in discussions about mathematical ideas effectively using math vocabulary.
 Differentiate Instruction	1.3.5 Collaboration offers a variety of entry points for different learning styles and can offer a variety of ways for students to engage with the activity. Use informal observation to determine which method best fits your classroom.	
	1.3.6 Your Turn! - students can post generated expressions in a shared space (e.g., class whiteboard or chart paper) using sticky notes or markers. Use this alternative process to check understanding and engage in discussion on identifying equivalent expressions since some questions have more than one correct answer. For students who finish 1.3.6 or 1.3.7 early, consider extending the Secret Number activity as a challenge to complete with a friend or family member at home.	
Lesson Notes:		

1.3.1 Warm-Up: Secret Number

Component Overview: Students use a series of clues related to number properties to identify a secret number. Consider having students think of their own number, then use similar math vocabulary to create clues for somebody else to identify their number.



1.3.1 Warm-Up: Secret Number

Juanita has a secret number. Read her clues and answer the questions that follow.

Clue 1: My secret number is a factor of 60.

1. What is the smallest value Juanita's number could be?
1
2. What is the largest value Juanita's number could be?
60

Clue 2: My number is prime.

3. What values are possible for Juanita's number at this point?
2, 3, or 5

Clue 3: 15 is a multiple of my secret number.
3 or 5

Clue 4: My secret number is a factor of 20.

4. What is Juanita's secret number?
5
5. Could you have determined her secret number only using the first three clues? Explain.
No, it only narrowed the answer down to 2 possibilities



Looking for an engaging idea for a project? Consider extending this activity by allowing students to write a series of interesting clues using factors, multiples, and other properties needed for somebody else to identify their secret number. This could also be an engaging homework activity.

1.3.2 Guided Instruction: The Distributive Property

Component Overview: Students are introduced to the definition of the distributive property and extend their understanding to addition expressions. Instruction also focuses on flexibility to rewrite expressions using different addends within the parentheses. Students also practice using expressions written with the distributive property to find products using mental math.



1.3.2 Guided Instruction: The Distributive Property



The **distributive property of multiplication over addition** is a property that applies to all rational numbers. The distributive property “distributes” a factor to all terms within a set of parentheses.

$$a \times (b + c) = (a \times b) + (a \times c)$$

The product 7×23 can be rewritten using the distributive property by decomposing 23 into two parts and multiplying 7 by both parts.

$$\begin{aligned} 7 \times 23 &= 7(20 + 3) \\ 7(20 + 3) &= (7 \cdot 20) + (7 \cdot 3) \end{aligned}$$

1. What are two other ways that 7×23 could be rewritten using the distributive property?

Answers will vary.

$$7 \times 23 = \underline{7} (\underline{10} + \underline{13})$$

$$7(23) = \underline{7} (\underline{11} + \underline{12})$$

2. Which of the expressions is the easiest to evaluate using mental math?

Answers will vary. $7(10 + 13)$; $7(20 + 3)$.

The distributive property can also be used to rewrite the sum of two numbers that share a common factor.



Consider building vocabulary and conceptual understanding of the distributive property by using the word “distribute” in a non-math context. Connect “distribute” to “give” with statements like, “If I distribute candy, that will mean EVERY person in this room would get a piece of candy. To distribute means to give.”



Have students practice mental math to find the product of an expression using the distributive property. For example, highlight that using the decomposed factors to multiply 7×20 and 7×3 , then adding the products, is easier than multiplying 7×23 mentally. Challenge students to generate another example and explain how mental math and distribution arrive at the same answer.

1.3.2 Guided Instruction: The Distributive Property (continued)

3. Elise rewrote the sum $27 + 18$ as $3(9 + 6)$. Are these two expressions equivalent?

- a. $27 + 18$ and $3(9 + 6)$ ☒ are ☐ are not equivalent

expressions because when evaluated they have

- ☒ the same value.
☐ different values.

- b. What common factor of 27 and 18 did Elise use to rewrite the expression?

3

- c. Rewrite the expression $27 + 18$ using a different common factor of 27 and 18. Use the area model to support your thinking.

	<u>3</u>	+	<u>2</u>
<u>9</u>	27		18

$$27 + 18 = \underline{9}(\underline{3} + \underline{2})$$



As an entry point for visual learners to “see” the distributive property, consider drawing arrows from the factor outside the parentheses to the addend inside the parentheses as you multiply to rewrite equivalent expressions as the sum of two products. Emphasize that the value is multiplied by both addends.



When rewriting sums, how can you determine which factor to distribute? Does the factor need to be the greatest common factor?

1.3.3 Guided Practice: Rewriting Expressions

Component Overview: Students synthesize learning of rewriting multiplication expressions in three ways: using the distributive property, modeling with area, and finding the product.



1.3.3 Guided Practice: Rewriting Expressions

- Complete the table by rewriting each multiplication expression using the distributive property, modeling with area, and finding the product.

Multiplication Expression	Distributive Property	Area Model	Product
98×5	$5(90 + 8)$	$ \begin{array}{r} 90 + 8 \\ 5 \begin{array}{ c c } \hline 450 & 40 \\ \hline \end{array} \end{array} $	490
6×22	<i>Answers will vary.</i> $6(12 + 10)$	$ \begin{array}{r} 12 + 10 \\ 6 \begin{array}{ c c } \hline 72 & 60 \\ \hline \end{array} \end{array} $	132

- Complete the table by rewriting each addition expression using the distributive property, modeling with area, and finding the sum.

Addition Expression	Distributive Property	Area Model	Sum
$600 + 140$	<i>Answers will vary.</i> $20(30 + 7)$	$ \begin{array}{r} 30 + 7 \\ 20 \begin{array}{ c c } \hline 600 & 140 \\ \hline \end{array} \end{array} $	740
$56 + 32$	<i>Answers will vary.</i> $8(7 + 4)$	$ \begin{array}{r} 7 + 4 \\ 8 \begin{array}{ c c } \hline 56 & 32 \\ \hline \end{array} \end{array} $	88



The second row of each table has multiple correct answers using various equivalent expressions. Encourage discourse on how various factors, not just the GCF, can be used to rewrite the expression using the distributive property. Use math vocabulary like “equal to” and “equivalent expressions” to highlight that the expressions are still equivalent.



When finding the product for these expressions, how can decomposing factors make it easier to multiply using mental math?



Consider an error analysis for the distributive property expressions in the question 2 table. For example, look at student work to identify a “My Favorite No.” Use this opportunity to emphasize procedural knowledge and a growth mindset by learning from mistakes.

1.3.4 Guided Instruction: Rewriting Sums Using Distributive Property

Component Overview: Students rewrite the sum of 16 and 24 using common factors and the distributive property, then analyze errors in rewritten expressions that are not equivalent. This activity provides a further opportunity for group or partner discussion to recognize, correct, and clarify procedural use of the distributive property to rewrite equivalent expressions.



1.3.4 Guided Instruction: Rewriting Sums Using Distributive Property

Sums of **composite whole numbers** can be rewritten using the distributive property and a common factor of the numbers.



A **composite number** is a whole number greater than 1 that has at least one whole-number factor other than one and itself.

- What are the common factors of 16 and 24?

Factors of 16	1, 2, 4, 8, 16
Factors of 24	1, 2, 3, 4, 6, 8, 12, 24
Common Factors of 16 and 24	1, 2, 4, 8

- Isaiah rewrote the sum $16 + 24$ using the distributive property. Each step in his thinking is shown. Complete the descriptions of Isaiah's process by examining his thinking and work. Use the words in the bank below to complete each description.

distributive equivalent factor multiplication

Isaiah's Thinking (💭) and Work (✍️)	Description
💭 "I know 2 is a common factor of both 16 and 24."	Isaiah identified a <u>factor</u> both numbers have in common.
✍️ $16 + 24 = 2(8) + 2(12)$	He rewrote the expression using <u>multiplication</u> with the common factor.
✍️ $2(8) + 2(12) = 2(8 + 12)$	He rewrote the expression using the <u>distributive</u> property.
💭 "The sum $16 + 24$ can be rewritten as $2(8 + 12)$ using the distributive property."	Isaiah recognized that $16 + 24$ and $2(8 + 12)$ are <u>equivalent</u> numeric expressions.



Consider activating prior knowledge from previous lessons:

- What is a factor? What is a common factor?
- Why would we need to be looking for common factors?
- Which factors can be used to create equivalent expressions?



Notice the icons used to represent different parts of the working process. Consider using these "thinking" and "work" prompts for your students to create additional opportunities for them to make sense of information.

1.3.4 Guided Instruction: Rewriting Sums Using Distributive Property (continued)

3. Rewrite the sum $16 + 24$ using the distributive property in two additional ways.

$$16 + 24 = 8(2 + 3)$$

$$16 + 24 = 4(4 + 6)$$

$2(8 + 12)$ is also a possible solution.

4. Two students made errors rewriting the sum $54 + 36$ using the distributive property. Each student's expression is shown.

Rewrite $54 + 36$ using the distributive property.	
Naveah's Expression	$4(14 + 9)$
Gavin's Expression	$3(18 + 36)$

Work with your partner to answer the following questions.

- What error did Naveah make?
 $4 \cdot 14 \neq 54$
 4 is not a common factor of 54 and 36 .
 - What error did Gavin make?
 $Gavin$ forgot to factor the 3 out of the 36 .
 - Generate two different equivalent expressions for the sum $54 + 36$ by rewriting it using the distributive property.
Answers will vary.
 $2(27 + 18)$
 $3(18 + 12)$
5. Rewrite an equivalent expression for the sum of $40 + 64$ using the distributive property with the **greatest common factor** (GCF).
 $8(5 + 8)$



The error analysis embedded within this activity gives students additional practice on MA.K12.MTR.4.1 and higher order thinking.



Clarify, Critique, and Correct

Consider emphasizing and asking students to identify and share reasons for why the errors lead to expressions that are not equivalent. Focus on reasoning connected to the correct use of the distributive property and identification of common factors.

1.3.5 Collaboration: Missing Values

Component Overview: Students work in partnerships to write missing values that make an equation true and write about how they determined the missing value. Consider having students verbalize reasoning for the values they used to make the equations true.



1.3.5 Collaboration: Missing Values

- Work with your partner to determine the missing values that will make each equation true.

a. $40 + 16 = 4(10 + 4)$ b. $15 + 80 = 5(3 + 16)$

c. $108 + 198 = 18(6 + 11)$ d. $28 + 63 = 7(4 + 9)$

e. $45 + 18 = 3(15 + 6)$ f. $135 + 27 = 9(15 + 3)$

- Choose one of the equations from question 1 and explain how you determined the missing value(s).
Answers will vary.

To find the missing value in equation _____, I
_____.

- Have your partner read your explanation and offer suggestions on how to improve it.



Determine how students are collaborating based on informal observations from the first few activities.

- Have students split the work (1a-c and 1d-f) and share their answers and reasoning.
- Have students work together, with one student only allowed to talk (progressing student) and other taking notes (proficient student).
- Have students choose which three to engage with (increasing choice and entry point) and find a partner around the room who did the same three to compare answers for questions 2 and 3.



Notice additional entry points using sentence stems in question 2 and intentional sharing for auditory learners in question 3. Consider using vocabulary cards to encourage students to replace words like “same number” with “common factor” when articulating their responses.

1.3.6 Your Turn!

Component Overview: Students work independently to rewrite expressions using common factors, including the GCF and the distributive property. Consider using this practice to encourage collaboration, flexible thinking, and discussion by allowing students to post generated equivalent expressions in a shared space.



1.3.6 Your Turn!

1. Rewrite an equivalent expression for each sum using the distributive property with a common factor.

a. $51 + 21$
 $3(17 + 7)$

b. $124 + 20$
 $2(62 + 10)$ or $4(31 + 5)$

c. $72 + 24$
 $24(3 + 1)$
 $12(6 + 4)$
 $8(9 + 3)$
 $6(12 + 4)$
 $4(18 + 6)$
 $3(24 + 8)$
 $2(36 + 12)$

d. $39 + 65$
 $13(3 + 5)$

2. Rewrite an equivalent expression for the sum of $68 + 34$ using the distributive property with the **greatest common factor** (GCF).

$34(2 + 1)$



Challenge students who finish early to create as many equivalent expressions as possible. They will notice that 1a and 1d only have one common factor, while option 1c has a variety.

1.3.7 Wrap-Up: Ticket Out the Door

Component Overview: Students apply their understanding to rewrite sums of two composite numbers using common factors and the distributive property.



1.3.7 Wrap-Up: Ticket Out the Door

1. Rewrite an equivalent expression for the sum of $75 + 30$ in two different ways using the distributive property and a common factor.

Answers will vary.

Possible answers include:

$15(5 + 2)$ or $5(15 + 6)$ or $3(25 + 10)$

2. Complete the equation with the missing values that make the equation true.

$$49 + 35 = 7 (7 + 5)$$



Consider supplementing these questions by asking students to answer one of the essential questions for this lesson in their own words. The essential questions are open-ended and allow students to represent their thinking in a format that best communicates their understanding. For example, one student could create a visual, step-by-step guide with labels explaining their response.

1.4 Multiplying Multi-Digit Numbers With Decimals to the Thousandths

Lesson Introduction

 Benchmark	MA.6.NSO.2.1 Multiply and divide positive multi-digit numbers with decimals to the thousandths, including using a standard algorithm with procedural fluency.		 Learning Target	I can fluently multiply multi-digit numbers with decimals to the thousandths.
 Benchmark Coherence	Building On MA.5.NSO.2.5 Multiply and divide a multi-digit number with decimals to the tenths by one-tenth and one-hundredth with procedural reliability.		Working Toward MA.7.NSO.2.2 Add, subtract, multiply, and divide rational numbers with procedural fluency.	
 Essential Questions	- How do you multiply decimals? - How is it helpful to think about place value and rewriting decimals as fractions when multiplying decimals? - What are the similarities and differences between multiplying decimals with the area model, partial products, and the standard algorithm?			
 Lesson Narrative	The lesson begins with an open-ended Warm-Up that reviews operations with decimals, followed by a Refresher that spirals learning from 5 th grade on reasoning with decimal operations. The lesson then continues by making connections to fractional thinking with base powers of 10, area models, partial products, and the standard algorithm for multiplying decimals to build conceptual understanding and procedural fluency.			
 Component Summary	1.4.1 Warm-Up (~5 min.)	Use place value reasoning to write factors with tenths (<i>SE p. 20</i>)	1.4.4 Guided Practice (5-10 min.)	Students use preferred method(s) to multiply decimals to the thousandths (<i>SE p. 23</i>)
	1.4.2 Refresher (5-10 min.)	Rewriting decimals as base-10 fractions (<i>SE p. 21</i>)	1.4.5 Try It (10 min.)	Students use preferred method(s) to multiply decimals to the thousandths (<i>SE p. 24</i>)
	1.4.3 Guided Instruction (10-15 min.)	Using area model, partial products, and standard algorithms (<i>SE p. 22</i>)	1.4.6 Your Turn! (5-10 min)	Students use any method(s) to multiply decimals to the thousandths (<i>SE p. 25</i>)
	1.4.7 Wrap-Up (~5 min.)	Fluency check for multiplying decimals (<i>SE p. 25</i>)		
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> - Area model - Partial product - Standard algorithm		(Optional) Place value chart (Optional) Chart paper (Optional) Markers	(Optional) Prepare chart paper and markers to create anchor charts showing all three methods introduced and practiced in 1.4.3 Guided Instruction and 1.4.4 Guided Practice. Consider posting in a shared classroom space as a reference on procedural steps to multiply decimals to refer back to after the lesson.

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may struggle with decimal place value, often confusing hundredths and thousandths. - Students may struggle with making sense of the standard algorithm. - Students may struggle with placement of the decimal in the product. Consider building understanding of base-10 thinking, fractional thinking, and estimation for scaffolding strategies as modeled in the 1.4.2 Refresher.	- This lesson's benchmark clarification calls for multiplying multi-digit numbers up to the thousandths with no more than 5 total digits, which are included in the lesson. However, not every problem goes to the thousandths as students are working the problems by hand. 1.4.2 Refresher is designed to make the connection between place value and powers of 10, but additional scaffolding may be needed for some students. - The standard algorithm is the most abstract method of solving. If you notice students struggling, connect it back to a more concrete method, such as area model, to build understanding. - If time is an issue, consider assigning all or part of 1.4.5 Your Turn! for homework.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Notice and Wonder 1.4.3 Guided Instruction and 1.4.4 Guided Practice give students opportunities to notice similarities and differences between the approaches for multiplying decimals presented in this lesson. Consider applying "Notice and Wonder" routines on each question set, positioning students to discover the relationship between partial products and standard algorithms.	MA.K12.MTR.3.1: Mathematical Fluency Throughout the lesson, students will be working to fluently multiply multi-digit decimals to the thousandths place using a variety of methods. Consider using 1.4.6 Your Turn! and 1.4.7 Wrap-Up components to provide plenty of practice using the standard algorithm, area model, and partial products to complete multiplication accurately and with confidence.
 Differentiate Instruction	1.4.6 Your Turn! - Consider specifying which problems students complete by complexity based on student confidence and comfortability with the standard algorithm. For students who are ready to multiply 5 digits, solve question 1a and 1c. For students who need additional practice with fewer digits, complete questions 1b and 1d. 1.4.3 Guided Instruction and 1.4.4 Guided Practice - Consider the area model as an entry point for visual learners and using arrows or color-coding the partial products within the area model to make specific connections to how they appear in the standard algorithm. 1.4.5 Try It – Consider having students create a math journal entry or interactive notebook page to solve each of three problems using one of the three methods. Consider labeling each method, creating an anchor poster on chart paper, and/or using this work as a visual reference for correctly used algorithms and models for students in later lessons.	

Lesson Notes:

1.4.1 Warm-Up: Decimal Products

Component Overview: Students use numbers 1 through 9 to write two factors to the tenths place that have a product as close to 50 as possible. Consider writing various student responses on a class whiteboard or shared space for an opportunity to review operations with decimals, place value, and rounding.



1.4.1 Warm-Up: Decimal Products

- Place a digit in each blue box so that the product is as close to 50 as possible. Use only digits 1 to 9, at most one time each.

$$\boxed{}.\boxed{} \times \boxed{}.\boxed{} =$$

Answers will vary.

Possible answers include: 6.1×8.2 ; 9.8×5.1



If students are unsure where to start, consider having students try a few different numbers that they created to notice how the values change. Use informal observations during this Warm-Up to determine entry points for students as they progress through the lesson.

1.4.2 Refresher: Reasoning With Powers of Ten

Component Overview: Students will be working in partners to reason with powers of 10 and fractional thinking as it relates to determining decimal place value when multiplying decimals. Consider using additional scaffolding using place value charts or other strategies to clearly make the connection between place value and powers of 10. The modeled thinking in this Refresher may also be helpful to revisit later in the lesson for any students who have difficulty placing decimals in products.



1.4.2 Refresher: Reasoning with Powers of Ten

Several expressions are shown.

$250 \cdot 0.01$	$250 \div 100$	$250 \cdot 0.001$	$250 \cdot \frac{1}{100}$	$250 \cdot 0.1$
$250 \div 1000$	$250 \cdot \frac{1}{10}$	$250 \cdot \frac{1}{1000}$	$250 \div 10$	

1. Work with your partner to complete the table by writing the expressions that match each value indicated.

Value	Equivalent Expressions		
25	$250 \cdot 0.1$	$250 \cdot \frac{1}{10}$	$250 \div 10$
2.5	$250 \cdot 0.01$	$250 \cdot \frac{1}{100}$	$250 \div 100$
0.25	$250 \cdot 0.001$	$250 \cdot \frac{1}{1000}$	$250 \div 1000$

2. When working together through the problem, Cheryl and Miranda had a disagreement.

Cheryl said that $\frac{1}{100}$ is equivalent to 0.001 because both have two zeros.	Miranda said that $\frac{1}{100}$ is equivalent to 0.01 because of place value.
--	---

- a. Which student do you agree with?
Answers will vary.
- b. Explain why you agree.
Answers will vary.



As an entry point for visual learners, consider writing the whole number factor and final product on a place value chart, and asking students to notice and wonder the difference between them.



For added complexity, consider asking students to provide a diagram, model, or other rationale as evidence for why they agree in question 2b. Options could include modeling fraction and decimal equivalence on a place value chart or using place value chips.

1.4.2 Refresher: Reasoning With Powers of Ten (continued)

Recall that fractions like $\frac{1}{10}$, $\frac{1}{100}$, and $\frac{1}{1000}$ can be used to find the location of the decimal point in terms of place value when multiplying decimals.

For example, 24×0.02 can be thought of as $24 \times \frac{2}{100}$ or $24 \times 2 \times \frac{1}{100}$. This is equal to $\frac{48}{100}$ which is read, "48 hundredths" and written as 0.48 in decimal form with the 8 in the hundredths place.

$$24 \times 0.02 = 0.48$$

3. With your partner, complete the equations and statements below to show how similar thinking can be used to find the products of decimals.

a.

$$(0.7) \cdot (0.09) = (7 \times \frac{1}{10}) \cdot (9 \times \frac{1}{100}) = (7 \cdot 9) \cdot (\frac{1}{10} \cdot \frac{1}{100}) = \boxed{63} \cdot \frac{1}{\boxed{1000}}$$

Written as a decimal, $(0.7) \cdot (0.09) = \underline{0.063}$.

b.

$$(1.5) \cdot (0.43) = (15 \times \frac{1}{10}) \cdot (43 \times \frac{1}{100}) = \boxed{645} \cdot \frac{1}{\boxed{1000}}$$

Written as a decimal, $(1.5) \cdot (0.43) = \underline{0.645}$.



Have students use correct vocabulary when naming decimals and fractions by place value to help strengthen connections for converting fractions and decimals. For example, for question 3a, ask students to read aloud 0.7 as "seven tenths," and explicitly make the connection to writing $\frac{7}{10}$ as a fraction.



Predicting estimations could be a helpful strategy for students' self-reflection and verifying solutions. Consider the following probing questions to position students to exercise these skills:

- *How does the size of the product compare to the size of the factor when the factor is multiplied by 0.1? 0.01? 0.001?*
- *Can you predict the outcome of multiplying 750 by 0.1 or 0.01 without calculating? If so, how?*

1.4.3 Guided Instruction: Multiplying Decimals to the Thousandths

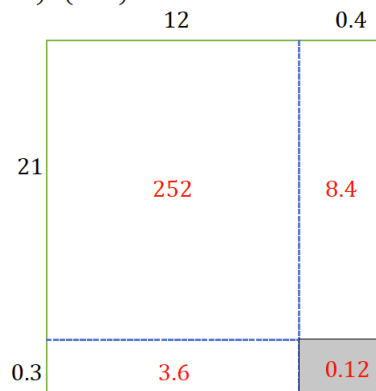
Component Overview: This section guides students through conceptual and procedural understanding to find the product of decimals to the thousandths place using three approaches: an area model, partial products, and the standard algorithm. The benchmark specifically identifies procedural fluency in the standard algorithm, but the lesson is designed to build on this appropriately based on prior learning.



1.4.3 Guided Instruction: Multiplying Decimals to the Thousandths

Different strategies can be used to find the product of decimals to the thousandths.

1. Byanca prefers using an area model to find the product of decimals. Below is an area model that Byanca drew to represent $(12.4) \cdot (21.3)$.



Consider asking students the following:

- Which parts of the factors would be multiplied to get 252?
- Which parts of the factors would be multiplied to get 8.4?

- a. Find the region that represents $(0.4) \cdot (0.3)$. Shade that region in the diagram and label it with its area of 0.12.
Answer shown in diagram.
- b. Label the other regions of the model with their respective areas.
Answer shown in diagram.
- c. Find the value of $(12.4) \cdot (21.3)$ using the labeled area model. Show your work.
 $252 + 8.4 + 3.6 + 0.12 = 264.12$

1.4.3 Guided Instruction: Multiplying Decimals to the Thousandths (continued)

2. Sergio prefers to use partial products to find the value of $(12.4) \cdot (21.3)$, while Alicia prefers using the standard algorithm. Their work is shown.

Sergio's Work Partial Products	Alicia's Work Standard Algorithm
$ \begin{array}{r} 12.4 \\ \times 21.3 \\ \hline 0.12 \\ 0.6 \\ 3 \\ 0.4 \\ 2 \\ 10 \\ 8 \\ 40 \\ + 200 \\ \hline 264.12 \end{array} $	$ \begin{array}{r} 12.4 \\ \times 21.3 \\ \hline 3.72 \\ 12.40 \\ + 248.00 \\ \hline 264.12 \end{array} $

Discuss each method with your partner while answering the following questions.

- a. In Sergio's work, which parts of the factors were multiplied to get each partial product indicated in the table below?

Partial Product	Parts of Factors Multiplied
0.12	<u>0.3</u> $\times$ <u>0.4</u>
0.6	<u>0.3</u> $\times$ <u>2</u>
3	<u>0.3</u> $\times$ <u>10</u>



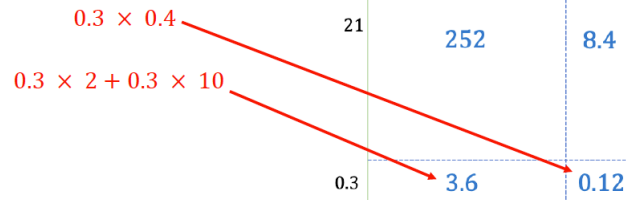
Notice and Wonder

Consider implementing a "Notice and Wonder" routine before students discuss and answer question 2.

- Looking at these two approaches, what do you notice is the same and different about the two approaches?
- What parts of the students' work is similar?
- Describe which method you think is more efficient.

1.4.3 Guided Instruction: Multiplying Decimals to the Thousandths (continued)

- b. To which parts of the area model do these partial products relate?



- c. Where do the values of 3.72, 12.4, and 248 come from in Alicia's work that shows the standard algorithm?

- 3.72 is the product that results from multiplying by each part of 12.4.
 - ☐ 0.3
 - ☐ 1
 - ☐ 20
 - 12.4 is the product that results from multiplying by each part of 12.4.
 - ☐ 0.3
 - ☐ 1
 - ☐ 20
 - 248 is the product that results from multiplying by each part of 12.4.
 - ☐ 0.3
 - ☐ 1
 - ☐ 20
- d. Draw lines to indicate connections between the partial products found in each of Sergio and Alicia's methods.
Answer shown in table.
- e. In each method, why are the numbers between the horizontal lines aligned vertically?
They are aligned vertically so the place values are aligned.



Using the nearest whole numbers of the factors, what is a good estimate for the product of these factors? How can I use this estimation to determine where to place the decimal point in the product of 12.4 and 21.3?



Notice that questions 2c and 2d position students to see explicit connections between the partial products method and the standard algorithm to build students understanding. Guide students to visual entry points using question 2d. As an entry point for auditory learners, verbalize the connection while drawing the arrows using phrases like "similar to" or "is the same as" to make connections.



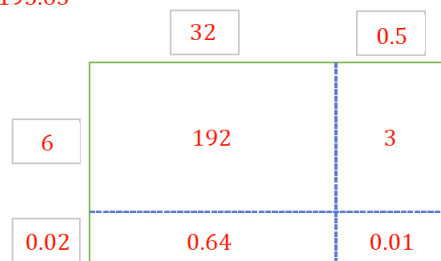
Consider the entry point for question 2d by helping draw the lines on question 2a, then ask students to draw the lines individually in question 2b. This support may be helpful for visual learners to first see the connection before identifying it on their own.

1.4.4 Guided Practice: Multiplying Decimals to the Thousandths

Component Overview: Students apply what they learned about multiplying decimals with the area model, partial products, and the standard algorithm to try each method with a partner. This practice guides students to identify errors, make corrections, and discuss students' opinions on the strengths/weaknesses of each method.

1.4.4 Guided Practice: Multiplying Decimals to the Thousandths

- Find the product $(32.5)(6.02)$ using the area model below. **195.65**



- With your partner, determine who will be Partner A and Partner B.
 - Find the product of 3.621×9.7 using the method indicated. Show your work in the space provided.

Partner A Partial Products	Partner B Standard Algorithm
$ \begin{array}{r} 3.621 \\ \times 9.7 \\ \hline 0.0007 \\ 0.014 \\ 0.42 \\ 2.1 \\ 0.009 \\ 0.18 \\ 15.4 \\ + 27 \\ \hline 35.1237 \end{array} $	$ \begin{array}{r} 3.621 \\ \times 9.7 \\ \hline 25347 \\ + 325890 \\ \hline 35.1237 \end{array} $

- Compare your work with your partner to verify that you both found the same product. Discuss any errors and make corrections as needed. Once you have come to an agreement, write your partner's work in your workbook in the space provided.
- Discuss with your partner the strengths and weaknesses of using each method.
- Which method do you prefer for multiplying decimals to the thousandths? **Answers will vary.**



Synthesis is critical at this stage. Consider how you are positioning students to navigate across methods and achieve fluency by selecting methods that work best for them.

- Intentionally include time for comparing and making sense in question 2
- Purposefully pair students of different learning styles together



If students are having difficulty with placement of decimal digits, avoid teaching shortcuts like counting the number of decimal places to determine the placement of the decimal in the product. Consider instead modeling fractional thinking or estimation to help students make conceptual sense of this.

1.4.5 Try It: Multiplying Decimals to the Thousandths

Component Overview: Students apply a preferred method to multiply decimals up to the thousandths place. Use this section as an opportunity to focus on procedural fluency. Consider having students that used different methods use their preferred method to solve one of the problems.



1.4.5 Try It: Multiplying Decimals to the Thousandths

1. Find each product using your preferred method. Show your work.

a. $25.36 \cdot 14.8$
 375.328

b. 19.008×3.2
 60.8256

c. $(36.3)(0.74)$
 26.862



Consider a math journal entry, foldable, or type of interactive notebook page for this practice. Students could solve each of the questions 1a-c using one of the three methods from this lesson, then check answers with partners or the teacher for accuracy. This could provide a record of correctly used methods for students to reference in later lessons.



For additional support both in 1.4.5 Try It and 1.4.6 Your Turn!, consider offering Math Nation's On-Ramp video on Multiplying Decimals to students.

1.4.6 Your Turn!

Component Overview: Students use any method to accurately multiply decimals to the thousandths place. Consider adapting by specifying which questions students attempt based on student needs and procedural fluency using various multiplication approaches. Questions 1a and 1c include factors with 5 digits, while questions 1b and 1d have factors with 2-4 digits.



1.4.6 Your Turn!

1. Find each product.

a. 105.46×3.8
400.748

b. $8.7 \cdot 0.475$
4.1325

c. $(7.7)(102.65)$
790.405

d. 1.561×12.8
19.9808



Consider using this section to encourage students to use more than one method to build fluency and conceptual understanding of decimal multiplication across approaches.



How can you use estimation to determine if your product is reasonable?

1.4.7 Wrap-Up: Ticket Out the Door

Component Overview: Consider using all or part of this fluency check as a formative assessment to gather data on student progress from this lesson. This Wrap-Up assesses students' ability to fluently multiply decimals to the thousandths place. It does not require the use of a particular method. Students should not use a calculator to complete the Wrap-Up.



1.4.7 Wrap-Up: Ticket Out the Door

1. Find each product.

a. $(6.36)(12.27)$
 78.0372

b. $8.12 \cdot 29.4$
 238.728

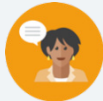


Encourage students to use the method of solving that makes the most sense to them. It may be helpful to provide a blank area model for students to fill in.

1.5 Dividing Multi-Digit Numbers With Decimals to the Thousandths

Lesson Introduction

 Benchmark	MA.6.NSO.2.1 Multiply and divide positive multi-digit numbers with decimals to the thousandths, including using a standard algorithm with procedural fluency.		 Learning Target	I can fluently divide multi-digit numbers with decimals to the thousandths.
 Benchmark Coherence	Building On: MA.5.NSO.2.5 Multiply and divide a multi-digit number with decimals to the tenths by one-tenth and one-hundredth with procedural reliability.		Working Toward: MA.7.NSO.2.2 Add, subtract, multiply, and divide rational numbers with procedural fluency.	
 Essential Questions	- How do you divide decimals using long division? - Why does multiplying the dividend and divisor by the same power of 10 not change the value of the quotient?			
 Lesson Narrative	The lesson begins with a Warm-Up and Refresher that reviews long division from 5 th grade and operations with decimals. The lesson then continues by exploring the relationship between the dividend, divisor, and quotient in long division with decimals. The lesson is designed for students to build understanding of the size of the quotient in terms of the dividend and divisor – not to teach shortcuts about decimal placement.			
 Component Summary	1.5.1 Warm-Up (~5 min.)	“Notice and Wonder” on decimal quotients (<i>SE p. 26</i>)	1.5.4 Bring It Together (5-10 min.)	Fill-in-the-blank activity to check conceptual understanding (<i>SE p. 28</i>)
	1.5.2 Refresher (~5 min.)	Review of long division with whole numbers (<i>SE p. 27</i>)	1.5.5 Guided Practice (5-10 min.)	Practice dividing decimals and recognizing how to multiply decimal values before dividing (<i>SE p. 29</i>)
	1.5.3 Exploration (15-20 min.)	Explore strategies for dividing decimals with values multiplied by powers of 10 (<i>SE p. 27</i>)	1.5.6 Your Turn! (5-10 min.)	Practice dividing decimals using long division (<i>SE p. 29</i>)
	1.5.7 Wrap-Up (~5 min.)	Reflection: Self-assessment of learning targets (<i>SE p. 30</i>)		
 Resources	Glossary		Materials	
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> - Quotient - Dividend - Divisor - Powers of 10		(Optional) Chart paper (Optional) Markers	
			Additional Preparation	
			N/A	

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may struggle with placement of the decimal in the quotient. - Students may confuse steps of the standard algorithm for long division.	- This lesson is built to make connections from students' prior knowledge of long division to now include decimals. Consider making explicit connections and highlighting how the decimal placement is determined. - To reinforce correct placement of the decimal in the quotient, consider building students' understanding of base-10 thinking using similar thinking as modeled in 1.5.3 Exploration.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
 Differentiate Instruction	Notice and Wonder 1.5.1 Warm-Up positions students to notice similarities and differences between values of quotients when dividing by decimals. Consider applying "Notice and Wonder" routines to encourage students' math thinking and help students to think about the size of the quotient in terms of the dividend and divisor.	MA.K12.MTR.3.1 Mathematical Fluency Throughout the lesson, students will be working to divide multi-digit decimals confidently and fluently to the thousandths place using long division. Consider using the 1.5.6 Your Turn! section to provide practice with generalizing how to multiply by powers of 10 to have a whole number divisor.
	1.5.6 Your Turn! – Consider specifying which problems students complete by difficulty based on student confidence and comfort with long division.	
	1.5.3 Exploration and 1.5.5 Guided Practice - Consider allowing partners to discuss reasoning and check each other's work on procedural accuracy with the long division algorithm. This partner work can also help auditory and visual learners hear and see different strategies and correct minor long division mistakes.	
1.5.5 Guided Practice - Consider having students create a math journal entry or interactive notebook page to solve each problem clearly with long division. Consider creating an anchor poster on chart paper and/or using this work as a visual reference for correct long division with decimals for students to refer to in later lessons.		
Lesson Notes:		

1.5.1 Warm-Up: Quotients as Decimals

Component Overview: Students use “Notice and Wonder” routines to look at quotients of several division expressions, all including 1 as the dividend and differing values as the divisor. Consider having students share in partners or as a class. Highlight responses that indicate how the size of the quotient changes in relation to the divisor’s value.



1.5.1 Warm-Up: Quotients as Decimals

The quotients of several division expressions are shown.

$1 \div 2 = 0.5$	$1 \div 3 = 0.333\overline{3}$	$1 \div 4 = 0.25$
$1 \div 5 = 0.2$	$1 \div 6 = 0.666\overline{6}$	$1 \div 7 = 0.14285\overline{7}$
$1 \div 8 = 0.125$	$1 \div 9 = 0.111\overline{1}$	$1 \div 10 = 0.1$
$1 \div 11 = 0.0909\overline{09}$	$1 \div 12 = 0.083\overline{3}$	$1 \div 13 = 0.07692\overline{3}$

- Write something you notice about the values of the quotients and something you still wonder.

I notice...	I wonder...
Answers will vary. I notice that the repeating decimals all have odd divisors. I notice that the quotients decrease as the divisor increases.	Answers will vary. I wonder if there is a way to predict repeating decimals.



Notice and Wonder

Ask the students to share what they noticed and wondered. Consider highlighting responses that focus on the size of the quotient changing in relation to the divisor as a primer for the lesson.

1.5.2 Refresher: Long Division

Component Overview: Students review the long division algorithm by filling in incomplete steps on a partially completed problem. Use this section as an opportunity to review procedural steps to divide using the long division algorithm and highlight important steps.



1.5.2 Refresher: Long Division

Jiang was dividing using long division and got stuck.

1. Complete the problem to show Jiang the missing steps.

$$\begin{array}{r}
 \begin{array}{c} \boxed{1} \end{array} \begin{array}{c} \boxed{2} \end{array} \begin{array}{c} \boxed{5} \end{array} \\
 15 \overline{) 1875} \\
 \underline{-15} \\
 \begin{array}{c} \boxed{3} \end{array} \begin{array}{c} \boxed{7} \end{array} \\
 - \begin{array}{c} \boxed{3} \end{array} \begin{array}{c} \boxed{0} \end{array} \\
 \underline{ \begin{array}{c} \boxed{7} \end{array} \begin{array}{c} \boxed{5} \end{array}} \\
 - \begin{array}{c} \boxed{7} \end{array} \begin{array}{c} \boxed{5} \end{array} \\
 \underline{ } \\
 0
 \end{array}$$

2. What advice about long division do you have that could help Jiang in the future?

Answers will vary.



Consider the purpose of this lesson when gauging time in this activity. Emphasize question 2 and the relationship between the divisor and dividend in relation to quotient, and utilize that same line of thinking when engaging with decimals.



Consider asking students to discuss this question rather than writing their answers. This entry point might engage auditory learners. Use sentence stems to encourage conversation such as, “One step Jiang needs to remember in the future is...” or “One detail that I have to remember when I divide is...”

1.5.3 Exploration: Dividing Decimals

Component Overview: Students work in partners to explore two strategies for finding the quotient of a decimal dividend and whole number divisor. Use this section to help students notice that multiplying the dividend and divisor by the same power of 10 does not change the value of the quotient, and to pay attention to the similarities and differences in the placement of the digits/decimal in the long division algorithm. Consider providing opportunities for students to share thinking and check long division work with partners.



1.5.3 Exploration: Dividing Decimals

Giovanny and Alina used different strategies to find the quotient of $48.78 \div 9$.

Giovanny's Strategy

$$\begin{array}{r} 5 \text{ } 4 \text{ } 2 \\ 9 \overline{) 48.78} \\ \underline{- 45} \\ 37 \\ \underline{- 36} \\ 18 \\ \underline{- 18} \\ 0 \end{array}$$

Alina's Strategy

$$\begin{array}{r} 5 \text{ } 4 \text{ } 2 \\ 900 \overline{) 4878} \\ \underline{- 4500} \\ 3780 \\ \underline{- 3600} \\ 1800 \\ \underline{- 1800} \\ 0 \end{array}$$

$$48.78 \div 9 = 5.42$$

Work with your partner to answer the questions about Giovanny and Alina's strategies.

- How are the two students' calculations alike and different?

Similarities	Differences
- Same quotient - Same non-zero digits in dividend and divisor	- Giovanny's strategy keeps the values as decimals - Alina multiplies the dividend and divisor by 100 first



Notice and Wonder

Prompt students to notice how the strategies and digits used are alike and different, specifically regarding the placement of digits and how their placement affects place value. Encourage discussion using sentence stems like "I notice the digit in Giovanny's strategy is...and the same digit in Alina's strategy is..." This attention to digits and place value will build conceptual understanding of decimal long division.



While circulating the room, monitor for students who are stuck or unsure what to look for when comparing.

- What is the same or different about the values Giovanny and Alina used as their divisor and dividend?
- What is the same or different about the quotient?
- What does the vertical line represent?
- How does the vertical line change in both strategies and what does this mean in relation to place value?

1.5.3 Exploration: Dividing Decimals (continued)

2. Look at Giovanni's calculations. Complete the statements with the appropriate place value to explain how you can tell that the 36 means "36 tenths" and the 18 means "18 hundredths" in his calculations.

- The 3 in 36 is in the ones place and the 6 is in the tenths place, so 36 in this calculation means 36 tenths.
- The 1 in 18 is in the tenths place, so it has the same value as 10 hundredths. The 8 is in the hundredths place. Together, they make 18 hundredths.

3. Look at Alina's calculations. Complete the statement to describe the values of 3600 and 1800 in her calculation.

- In Alina's calculations, the 3600 represents 3600 tenths, and the 1800 represents 1800 hundredths.

4. Why is the value of $48.78 \div 9$ the same as the value of $4878 \div 900$?

The numbers in the second expression are both multiplied by 100, so the value of the quotient will not change.

5. Explain why $51.84 \div 6.4$ has the same value as $518.4 \div 64$.

The numbers in the second expression are both multiplied by 10, so the value of the quotient will remain the same.

6. Find the value of $51.84 \div 6.4$ using long division. Show your work.

$$\begin{array}{r} 8.1 \\ 6.4 \overline{) 51.84} \\ \underline{- 51.2} \\ 0.64 \\ \underline{- 0.64} \\ 0 \end{array}$$



Encourage students to complete the statements while also verbally explaining their reasoning as an entry point for auditory learners.



Consider probing questions to encourage synthesis for students as they formalize their own approach to dividing with decimals:

- What are some advantages of calculating the quotient with the decimals intact, like Giovanni?*
- What are some advantages of rewriting the dividend and divisor as whole numbers before dividing, like Alina?*
- Why does multiplying the dividend and divisor by the same power of 10 not change the value of the quotient?*

1.5.4 Bring It Together: Dividing Decimals

Component Overview: Students complete statements to confirm conceptual understanding of why the value of a quotient does not change when both the divisor and dividend are multiplied by the same power of 10. Consider allowing students to complete the statements individually first, then check for correctness and add complexity by having students provide an example as evidence.



1.5.4 Bring It Together: Dividing Decimals

1. Complete the summary of the exploration of dividing decimals using the words and phrases from the bank below.

does not change	multiply
power of 10	whole numbers

One way to find a quotient of two decimals is to first

multiply each decimal by a power of 10 so that both values are whole numbers. If we multiply both values by the same power of 10, this does not change the value of the quotient.



Consider adding complexity by having students write their own example of two division statements whose quotient is the same that have different decimal divisors and dividends. Students can write their examples in the space below and have a partner confirm their division statements are equivalent.

1.5.5 Guided Practice: Dividing Decimals

Component Overview: Students practice dividing decimals by first multiplying the divisor and dividend by a power of 10. Consider emphasizing conceptual understanding and procedural fluency by having students solve problems using various versions of the divisor and dividend that will have the same quotient.



1.5.5 Guided Practice: Dividing Decimals

1. Find each quotient using a strategy similar to Giovanni's or Alina's. Show your work.

a. $24.2 \div 1.1$
22

b. $17.028 \div 0.08$
212.85

2. Coby and Olivia each found the quotient of $10.92 \div 2.6$. The values they each used are listed.

- Coby calculated $1092 \div 260$.
- Olivia calculated $109.2 \div 26$.

- a. Will the values they used result in the same quotient as $10.92 \div 2.6$? Explain.
Yes, Coby multiplied both values by 100 first and Olivia multiplied both values by 10 first.

b. Find the quotient of $10.92 \div 2.6$.
4.2



Consider probing questions for students who are stuck that align with the benchmark for this lesson:

- If you need a whole number divisor before dividing, how do you determine which power of 10 to multiply both decimals by?
- Which power of 10 will you use for question 1a? For question 1b? Why?



Notice the skills and higher order thinking necessary for engaging students in question 2. While monitoring during this practice time, consider highlighting two differing responses from students and have them share their thinking with the class.

1.5.6 Your Turn!

Component Overview: Students apply learning in this section by dividing decimals using any strategy. Consider choosing which problems students work on based on learner preparedness and using a math journal or interactive notebook page to record strategies as a reference for students in later lessons.



1.5.6 Your Turn!

1. Find each quotient. Show your work.

a. $43.396 \div 57.1$
0.76

b. $29.88 \div 0.36$
83



Consider using specific problems from this section for different learner fluency with divisibility rules. For example, questions 1a and 1d include multiplying by 1,000 and therefore requires students to think about divisibility with larger values. Questions 1b and 1c involve multiplying by 100 and therefore includes smaller whole values to divide.

c. $262.5 \div 5.25$
50

d. $15.129 \div 12.3$
1.23

1.5.7 Wrap-Up: Reflection

Component Overview: This self-assessment asks students to consider their procedural understanding and comfort with the lesson's learning target. Encourage students to revisit work throughout the lesson as evidence of learning.



1.5.7 Wrap-Up: Reflection

1. Draw an **X** on the line to indicate where your current learning is in relation to the learning target.

Answers will vary.

I can fluently divide multi-digit numbers with decimals to the thousandths.

I need help.
I can't get started.

I'm getting there.

I understand
and have evidence.



Consider pairing the “number” line responses seen here with informal observations throughout the lesson for intentional grouping in Lesson 7 that targets division of fractions.

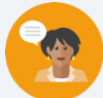
2. Briefly explain why you placed the X where you did.

Answers will vary.

1.6 Multiplying Positive Fractions

Lesson Introduction

 Benchmark	MA.6.NSO.2.2 Extend previous understanding of multiplication and division to compute products and quotients of positive fractions by positive fractions with procedural fluency.		 Learning Target	I can fluently multiply positive fractions by positive fractions.	
 Benchmark Coherence	Building On		Working Toward		
	MA.5.FR.2.2 Extend previous understanding of multiplication to multiply a fraction by a fraction, including mixed numbers and fractions greater than 1, with procedural reliability.		MA.7.NSO.2.2 Add, subtract, multiply, and divide rational numbers with procedural fluency.		
 Essential Questions	- How do you multiply fractions? - What strategies can be used to multiply fractions? - How can fractions greater than 1 be rewritten?				
 Lesson Narrative	This lesson begins with a Career Profile Video Warm-Up, followed by a Refresher on multiplying fractions that spirals multiplying fractions from 5 th Grade and using GCF to simplify fractions from Lesson 1. The Refresher leads students through multiplying fractions using the standard algorithm rather than the area model. The rest of the lesson helps students work toward procedural fluency with the standard algorithm and practice a variety of problems.				
 Component Summary	1.6.1 Warm-Up (~5-10 min)	Career profile video (<i>SE p. 31</i>)	1.6.4 Try It (10-15 min)	Practice multiplying fractions and reflecting on strategies used (<i>SE p. 32</i>)	
	1.6.2 Refresher (10-15 min)	Review multiplying fractions in pairs (<i>SE p. 31</i>)	1.6.5 Your Turn! (10 min)	Problem set to find products of mixed numbers and fractions (<i>SE p. 33</i>)	
	1.6.3 Guided Instruction (5-10 min)	Step-by-step practice multiplying fractions and rewriting in simplest form using GCF (<i>SE p. 32</i>)	1.6.6 Wrap-Up (~5 min)	Spotlight Reflection to self-assess concepts from unit (<i>SE p. 33</i>)	
 Resources	Glossary		Materials		Additional Preparation
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> - Mixed number - Greatest common factor (GCF) - Simplest form		Career Profile Video “Food Science”		1.6.1 Warm-Up contains a video to accompany the question prompts.

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may struggle to recognizing equivalent fractions. - Students may multiply when converting from an improper fraction to a mixed number. - Students may not distribute all parts of the whole number and the fraction when multiplying.	- Some students may need more scaffolding to recall how to convert between mixed numbers to fractions greater than 1, or vice versa. - Consider making connections between the numerators and denominators of the factors and how this can be used to rewrite the fraction in an equivalent form. Help students recognize equivalent forms are all correct, but the preciseness of simplifying may be required. - Consider explicitly showing how distribution happens when multiplying mixed-number fractions by using the area model to make connections (as modeled in 1.6.2 Refresher).
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Number Talk 1.6.4 Try It – Consider using Number Talk routines to allow students to think about and attempt to solve a problem individually, then share ideas and strategies as a group. This instructional routine fosters flexibility and procedural fluency. Encourage students to be precise in their word choice and use of language when describing how they found their answer.	MA.K12.MTR.5.1: Patterns and Structure 1.6.2 Refresher and 1.6.3 Guided Instruction position students to review previously learned concepts of fraction multiplication, the area model, and using the GCF to rewrite fractions in simplest form. Consider pointing out how these concepts connect to this lesson’s targets, including how the area model and rewriting fractions connects to algorithmic procedures for multiplying fractions.
 Differentiate Instruction	1.6.5 Your Turn! - Consider placing students in groups and assigning groups to solve certain Your Turn! problems. By differentiating which groups engage with specific problems, this section of the lesson can become a jigsaw in which students are responsible for sharing their approach and answers with others.	
	1.6.2 Refresher provides a visual entry point by connecting the algorithm to multiplying mixed number fractions with the area model. Consider having students use area models as they make connections to how the distributive property is used.	
	1.6.6 Wrap-Up positions students to think about their learning through “Stoplight Reflections” for important operational concepts from this unit. Consider a variety of methods for displaying and interpreting these reflections.	
Lesson Notes:		

1.6.1 Warm-Up: Food Science

Component Overview: Display the “Career Profile – Food Science” video for the entire class. After students watch this video, prompt the students to answer the question independently. Consider giving students an opportunity to share their responses with classmates.



1.6.1 Warm-Up: Food Science

Dr. Charlotte Armah is a research scientist at the Institute of Food Research. She leads experiments involving human volunteers to learn whether or not eating particular foods can protect us from diseases such as cardiovascular disease and cancer. Her work allows her to combine her scientific and math expertise, interest in helping others, and enjoyment of interacting with different people.



Do you know someone who is a research scientist or works in food science? Consider bringing in a visitor for your class to ask questions and learn from. Additionally, consider having students return to this lesson later in the year for project-based learning around this Career Profile.

1. What is one aspect of Charlotte’s story that was surprising to you? Explain why this stood out to you.

Answers will vary.

1.6.2 Refresher: Multiplying Fractions

Component Overview: Students work in partners to review how to multiply fractions. This section helps connect the algorithmic approach with the area model and with rewriting mixed numbers as a fraction greater than 1 before multiplying. Though the area model is used as a scaffold and to make connections, be sure to reinforce procedural steps using the standard algorithm to achieve algorithmic fluency.

1.6.2 Refresher: Multiplying Fractions



When **multiplying fractions**, multiply the numerators of the factors to find the numerator of the product, and multiply the denominators of the factors to find the denominator of the product.

$$\frac{3}{4} \times \frac{2}{5} = \frac{6}{20}$$

1. With your partner, determine who will be partner A and who will be partner B.
 - a. Find the product $1\frac{2}{3} \times 3\frac{8}{9}$ using the method indicated.

Partner A: Area Model	Partner B: Rewrite the Mixed Number(s) as a Fraction Greater than 1
$3 + 2 + \frac{8}{9} + \frac{16}{27} = 6\frac{13}{27}$	$1\frac{2}{3} \times 3\frac{8}{9}$ $\frac{5}{3} \cdot \frac{35}{9} = \frac{175}{27} = 6\frac{13}{27}$

- a. Compare your work with your partner to verify that you both found the same product. Discuss any errors and make corrections as needed. Once you are in agreement, write your partner's work in your workbook in the space provided.



Consider asking students the following:

- Are $6\frac{13}{27}$ and $\frac{175}{27}$ equivalent?
- Do products found as fractions greater than 1 always need to be rewritten as mixed numbers?



After students have time to compare strategies, ask if the same product could be attained by multiplying the whole number parts by each other and the fractional parts by each other and then combining. Ensure students understand the distribution that is happening, and which parts are being multiplied using the area model to make connections.

1.6.3 Guided Instruction: Multiplying Fractions

Component Overview: This section guides students through multiplying fractions step-by-step, including first rewriting mixed numbers and multiplying, then using the GCF of the numerator and denominator to write an equivalent fraction. Consider completing the first problem together as a class or in partners to review how to use the GCF to rewrite fractions in simplest form from Lesson 1.



1.6.3 Guided Instruction: Multiplying Fractions

Recall from Lesson 1 that fractions can be rewritten in simplest form using the GCF of the numerator and denominator.

Find each product using the steps indicated.

1.

$$\frac{5}{8} \times 4\frac{2}{3}$$

Write the given multiplication expression.	$\frac{5}{8} \cdot 4\frac{2}{3}$
Rewrite the expression, rewriting any mixed numbers as fractions greater than 1.	$\frac{5}{8} \cdot \frac{14}{3}$
Multiply to find the product of the fractions.	$\frac{70}{24}$
Rewrite the product in simplest form using the GCF of the numerator and denominator.	$\frac{35}{12}$ or $2\frac{11}{12}$

2.

$$3\frac{3}{5} \cdot 2\frac{1}{6}$$

Write the given multiplication expression.	$3\frac{3}{5} \cdot 2\frac{1}{6}$
Rewrite the expression, rewriting any mixed numbers as fractions greater than 1.	$\frac{18}{5} \cdot \frac{13}{6}$
Multiply to find the product of the fractions.	$\frac{234}{30}$
Rewrite the product in simplest form using the GCF of the numerator and denominator.	$\frac{39}{5}$ or $7\frac{4}{5}$



Encourage recall of concepts and math vocabulary from Lesson 1 prior to starting the problems in this section. Questions could include: “What is the GCF? How do we find the GCF? How is the GCF used to write a fraction in an equivalent form?”



To strengthen understanding for visual learners, consider having students draw arrows from each step in the left column to their work on the right column. This can help students make sure they completed each step, connect a written step to a visual example, and encourage procedural understanding on how to multiply fractions step-by-step.



For question 2, ask students, “How can you determine that the product will be able to be simplified when looking at the factors multiplied?” Students may offer multiple methods, some of which do not include the greatest common factor. Build fluency (MA.K12.MTR.3.1) by emphasizing the usefulness of the greatest common factor.

1.6.4 Try It: Multiplying Fractions

Component Overview: Students have four questions to practice multiplying fractions and rewrite in simplest form. Consider using any part of question 1 for Number Talk routines and practicing procedural fluency, then question 2 to position students to reflect on which questions were most challenging and on preferred strategies for future problems.



1.6.4 Try It: Multiplying Fractions

- Find each product. Rewrite each product in simplest form. Show your work.

a. $\left(1\frac{1}{8}\right)\left(5\frac{5}{7}\right)$
 $\frac{9}{8} \cdot \frac{40}{7} = \frac{360}{56} = \frac{45}{7}$ or $6\frac{3}{7}$

b. $4\frac{4}{5} \times \frac{10}{13}$
 $\frac{24}{5} \cdot \frac{10}{13} = \frac{240}{65} = \frac{48}{13}$ or $3\frac{9}{13}$

c. $\frac{4}{11} \cdot \frac{1}{2}$
 $\frac{4}{11} \cdot \frac{1}{2} = \frac{4}{22} = \frac{2}{11}$

d. $2\frac{1}{3} \times 4\frac{4}{5}$
 $\frac{7}{3} \cdot \frac{24}{5} = \frac{168}{15} = \frac{56}{5}$ or $11\frac{1}{5}$

- Complete the statement to describe which product was the most challenging for you to find.

Answers will vary.

The product of _____ was most challenging for me to find because _____.

- What is a strategy that could help you on a similar problem in the future?

Answers will vary.



Number Talk

Consider using any of the questions 1a-d as an opportunity to engage in “Number Talk” routines.

Pick one problem for students to consider different approaches to solving, then share those ideas as a group.

Highlight responses that:

- Rewrite mixed numbers to simplify or reduce before multiplying (fluency).
- Utilize both GCF and other common factors.
- Utilize precision in math vocabulary.
- Show different “correct” answers to compare more precise responses, like $\frac{4}{22}$ or $\frac{2}{11}$ in 1c.

1.6.5 Your Turn!

Component Overview: Students independently practice finding products of mixed numbers and fractions and simplifying when possible. Consider having students work in groups to solve three problems, then share answers and strategies in a jigsaw activity.



1.6.5 Your Turn!

1. Find each product. Simplify when possible.

a. $\frac{2}{3} \times \frac{5}{4}$
 $\frac{10}{12} = \frac{5}{6}$

b. $2\frac{1}{3} \times \frac{3}{4}$
 $\frac{7}{3} \cdot \frac{3}{4} = \frac{21}{12} =$
 $\frac{7}{4}$ or $1\frac{3}{4}$

c. $\frac{8}{7} \cdot \frac{7}{10}$
 $\frac{56}{70} = \frac{4}{5}$

d. $\frac{13}{2} \cdot \frac{7}{4}$
 $\frac{91}{8}$ or $11\frac{3}{8}$

e. $7\frac{5}{6} \times 9$
 $\frac{47}{6} \cdot \frac{9}{1} = \frac{423}{6} =$
 $\frac{141}{2}$ or $70\frac{1}{2}$

f. $\left(\frac{24}{27}\right)\left(\frac{45}{32}\right)$
 $\frac{8}{9} \cdot \frac{5}{8} = \frac{5}{4}$

g. $\frac{3}{2} \cdot \frac{7}{10}$
 $\frac{21}{20}$ or $1\frac{1}{20}$

h. $\left(2\frac{6}{25}\right)\left(\frac{1}{25}\right)$
 $\frac{56}{25} \cdot \frac{1}{25} = \frac{56}{625}$

i. $4\frac{1}{5} \times 1\frac{7}{9}$
 $\frac{21}{5} \cdot \frac{16}{9} = \frac{336}{45} =$
 $\frac{112}{15}$ or $7\frac{7}{15}$



Students should independently engage with problems for meaningful practice.

If more collaboration is necessary depending on the level of your students, consider splitting the class into groups and assigning each group three problems to complete, then share how they solved each problem with the other two groups. Note that questions 1b, 1e, and 1h include mixed numbers, and all students should independently engage with at least two of these questions.

This jigsaw activity will provide an opportunity for students to share strategies and work on precise and accurate math language as they describe their strategy and share their answers with the other groups.

1.6.6 Wrap-Up: Reflection

Component Overview: Students use the “Stoplight Reflection” format to self-assess learning on important concepts from this unit. Consider having students write responses, share how they are feeling verbally to a partner, or share their reflection with the teacher individually.



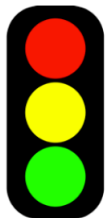
1.6.6 Wrap-Up: Reflection

In this unit, you have learned multiple concepts including those listed below.

- Finding the GCF and LCM of two or more numbers
- Rewriting sums using distributive property
- Multiplying and dividing decimals
- Multiplying fractions

1. Use the stoplight reflection below to share how you are feeling about each of these concepts at this point.

Answers will vary.



I still feel stuck about how to...

I’m starting to get the hang of...

I feel confident and ready to go with...



Notice the inclusion of concept stems for students to use as additional entry points for engaging in self-reflection.

- Consider providing an example of each concept for students to use as an anchor for reflection.
- Consider providing each student with four mini-flag sticky notes for each concept and have chart paper displayed with the stoplight. This visual representation will give you an aggregate of all students’ reflections.

1.7 Dividing Positive Fractions

Lesson Introduction

 Benchmark	MA.6.NSO.2.2 Extend previous understanding of multiplication and division to compute products and quotients of positive fractions by positive fractions with procedural fluency.		 Learning Target	I can fluently divide positive fractions by positive fractions.
 Benchmark Coherence	Building On MA.5.FR.2.2 Extend previous understanding of multiplication to multiply a fraction by a fraction, including mixed numbers and fractions greater than 1, with procedural reliability.		Working Toward MA.7.NSO.2.2 Add, subtract, multiply, and divide rational numbers with procedural fluency.	
 Essential Questions	- How do you divide fractions using the standard algorithm? - How can a tape diagram represent fraction division? - How can you connect models of division and multiplication to the standard algorithm?			
 Lesson Narrative	Tape diagrams are used to model division with fractions before moving to the standard algorithm. Students activate their prior knowledge and use of tape diagrams with division to model the same process for division of fractions. Students begin to formalize their tape model diagrams as standard algorithms.			
 Component Summary	1.7.1 Warm-Up (~5 min.)	Multiplying fractions Error Analysis (<i>SE p. 34</i>)	1.7.3 Guided Practice (10-15 min.)	Practice using any method to divide fractions (<i>SE p. 37</i>)
	1.7.2 Guided Instruction (15-20 min.)	Scaffolded practice dividing fractions with tape diagram and standard algorithm (<i>SE p. 35</i>)	1.7.4 Your Turn! (5-10 min.)	Students apply any method to divide fractions (<i>SE p. 38</i>)
	1.7.5 Wrap-Up (~10 min.)	Practice multiplying and dividing fractions with procedural fluency (<i>SE p. 38</i>)		
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> - Reciprocal - Existence of multiplicative inverses <u>Additional Terms:</u> - Mixed number		(Optional) Paper or cardstock (Optional) Markers	If using station rotations or "Scavenger hunt" activity for 1.7.4 Your Turn!, be sure to write each of the problems clearly on a piece of paper or cardstock and place throughout the room before the lesson.

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may forget to multiply by the reciprocal or incorrectly find the reciprocal when using the standard algorithm. - Students may confuse or switch the divisor and dividend fractions when computing. - Students may struggle to recognize when they can simplify.	- Consider scaffolding creation of tape diagrams to ensure student work accurately represents division expressions. - During 1.7.2 Guided Instruction and 1.7.3 Guided Practice, make connections between numerators and denominators of factors and how this can be used to recognize when products will be able to be simplified.
	Literacy and Language Routines Think-Pair-Share 1.7.1 Warm-Up positions students to analyze an error made when multiplying mixed numbers. Consider applying Think-Pair-Share routines as students identify an error in the algorithm used, positioning students to recognize and share a common mistake and provide feedback on how to avoid the mistake going forward.	Mathematical Thinking and Reasoning (MTR) Standard MA.K12.MTR.2.1: Multiple Representations Students will be working to model fraction division using tape diagrams. This standard emphasizes progressing from modeling problems with diagrams to using algorithms and equations – utilize the progression of the lesson to highlight this progress as students move from the tape diagram to using the standard algorithm.
 Differentiate Instruction	1.7.1 Warm-Up – consider having students write their error analysis notes on a sticky note, including diagrams with labels showing the error and/or correct work, and post these notes in a shared classroom space for students to learn from mistakes and foster a growth mindset in mathematics.	
	1.7.2 Guided Instruction – adjust the tape diagram creation to show fraction division more clearly with color or shading. Consider using colored pencils to circle or shade the dividend into groups that are the size of the divisor.	
	1.7.4 Your Turn! – consider trying ideas like stations, scavenger hunts, problem cards, or relay races to add variety and movement to student practice.	
Lesson Notes:		

1.7.1 Warm-Up: Error Analysis

Component Overview: Students analyze an error made when multiplying fractions and write a brief note to explain the mistake. Consider having students engage in a Think-Pair-Share for this activity.



1.7.1 Warm-Up: Error Analysis

Mateo tried to multiply $3\frac{5}{6} \cdot 4\frac{1}{10}$, but he made an error. His work is shown.

$$3\frac{5}{6} \cdot 4\frac{1}{10} = 12\frac{5}{60} = \left(12\frac{1}{12}\right)$$

- What error did Mateo make?
He did not convert the mixed fractions to improper fractions prior to multiplying.
- What is something that Mateo did correctly?
He correctly simplified the fraction.
- Help Mateo understand and learn from his mistake. Write a brief note to Mateo explaining the mistake he made and how to fix it moving forward.

Means
I
Start
To
Acquire
Knowledge
Experience
Skills

Answers will vary.



Think-Pair-Share

Consider having students work on the error analysis individually, then share their responses with a partner. Once the partner conversations have taken place, ask a few students to share their thoughts with the class. Also consider having students write on sticky notes to post in a shared classroom space, which could foster a growth mindset and a classroom environment that values learning from mistakes.

- Find the product of $3\frac{5}{6} \cdot 4\frac{1}{10}$. Show your work.

$$\frac{23}{6} \cdot \frac{41}{10} = \frac{943}{60} = 15\frac{43}{60}$$

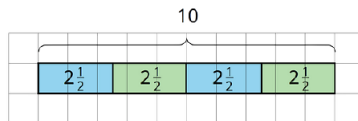
1.7.2 Guided Instruction: Dividing Fractions

Component Overview: Students use a tape diagram to illustrate division and find quotients with fractions. Complete the tape diagrams in the first section as a class to address any misconceptions and build conceptual understanding of dividing fractions. Consider having students work with a partner to apply their understanding to a real-world problem. Finally, students use the standard algorithm to divide fractions. Consider helping students see the reason behind why multiplying by the reciprocal in the standard algorithm is the same as dividing.



1.7.2 Guided Instruction: Dividing Fractions

Recall how tape diagrams can be used to illustrate division.



- Write the division equation represented by the tape diagram.

$$\begin{array}{ccccc} \text{dividend} & \div & \text{divisor} & = & \text{quotient} \\ 10 & \div & 2\frac{1}{2} & = & 4 \end{array}$$

- How many groups of $2\frac{1}{2}$ are in 10?
4

- Complete the table by finding each quotient. Use the tape diagram shown.

Tape Diagram	Division Expression	Quotient
	$\frac{1}{2} \div \frac{1}{3}$	$\frac{3}{2}$ or $1\frac{1}{2}$
	$3\frac{1}{4} \div \frac{3}{4}$	$\frac{13}{3}$ or $4\frac{1}{3}$



Throughout this guided instruction, consider using the sentence stem “How many groups of ___ are in ___?” This will reinforce student understanding of the divisor being the group size and the dividend being the total value that is divided into groups.



How many groups of $\frac{1}{3}$ are in $\frac{1}{2}$? How many groups of $\frac{3}{4}$ are in $3\frac{1}{4}$? How can you “see” the quotient to each problem in the tape diagram?



For a visual entry point, consider using colored pencils to show the dividend being divided by the divisor. Use different colors to represent each and circle each group to show how many groups of the divisor are in the dividend.

1.7.2 Guided Instruction: Dividing Fractions (continued)

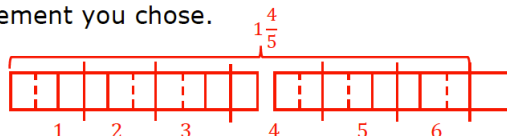
4. Jacquelyn is making ground beef flautas, where each flauta will use $\frac{3}{10}$ pounds of ground beef. She has a total of $1\frac{4}{5}$ pounds of ground beef to use.

- a. Which division statement below represents the number of flautas Jacquelyn can make?

$$\frac{3}{10} \div 1\frac{4}{5}$$

$$1\frac{4}{5} \div \frac{3}{10}$$

- b. Draw a tape diagram to illustrate the division statement you chose.



- c. Solve the division statement.

$$1\frac{4}{5} \div \frac{3}{10} = 6$$

- d. How many $\frac{3}{10}$ pound flautas can she make?

6 flautas

You previously developed an algorithm to use when dividing fractions.



When **dividing fractions**, the division statement can be rewritten as an equivalent multiplication sentence by using the **reciprocal**, or multiplicative inverse, of the second fraction.

$$\frac{3}{4} \div \frac{2}{5} = \frac{3}{4} \times \frac{5}{2} = \frac{15}{8} = 1\frac{7}{8}$$



Existence of multiplicative inverses

For every $a \neq 0$ there exists $\frac{1}{a}$ so that $a \times \frac{1}{a} = 1$ and $\frac{1}{a} \times a = 1$.



Think-Pair-Share

Consider having students complete this real-world fraction division problem using the Think-Pair-Share routine. Be sure to give students individual thinking time before allowing partners to share ideas and answers, then have partners share ideas with the class. Use this opportunity to address any misconceptions when dividing using a tape diagram for the whole class.



For question 4b, how do you use the dividend from your chosen division statement to start creating your tape diagram? How do you know how to divide the tape diagram? For questions 4c and 4d, how can you “see” the quotient in your tape diagram?

1.7.2 Guided Instruction: Dividing Fractions (continued)

With the tape diagrams, you determined $3\frac{1}{4} \div \frac{3}{4} = 4\frac{1}{3}$.

The standard algorithm can also be used to determine the quotient.

Write the given division expression.	$3\frac{1}{4} \div \frac{3}{4}$
Rewrite the expression, rewriting any mixed numbers as fractions greater than 1.	$\frac{13}{4} \div \frac{3}{4}$
Rewrite the division using multiplication by the reciprocal of the second fraction.	$\frac{13}{4} \times \frac{4}{3}$
Multiply the fractions and simplify.	$\frac{52}{12} = \frac{13}{3} = 4\frac{1}{3}$

5. Find the quotient $5\frac{2}{3} \div 2\frac{4}{9}$ using the algorithm.

$$\frac{17}{3} \div \frac{22}{9} = \frac{17}{3} \cdot \frac{9}{22} = \frac{153}{66} = \frac{51}{22} \text{ or } 2\frac{7}{22}$$



Encourage students to formulate their own steps, then discuss the steps in the table. Relate the steps to the tape diagram.



Consider using the algorithm as an opportunity to dive deeper into why multiplying by the reciprocal is the same as dividing. For example, provide fraction bar or tape diagram evidence that shows $10 \div 2 = 5$, and $10 \times \frac{1}{2} = 5$.

1.7.3 Guided Practice: Dividing Fractions

Component Overview: Students apply any method to divide fractions. The table in this section summarizes the important skills of rewriting mixed numbers as fractions before computing, writing the equivalent multiplication sentence with the reciprocal, and simplifying the quotient if possible. Consider giving students time to work individually, then checking responses as a class to confirm understanding and correct usage of the algorithm.



1.7.3 Guided Practice: Dividing Fractions

1. Complete the table to find each quotient.

Division Sentence	Rewritten as Fractions (If needed)	Equivalent Multiplication Sentence	Quotient
$\frac{7}{8} \div 2\frac{3}{10}$	$\frac{7}{8} \div \frac{23}{10}$	$\frac{7}{8} \cdot \frac{10}{23}$	$\frac{70}{184} = \frac{35}{92}$
$\frac{2}{7} \div \frac{3}{4}$		$\frac{2}{7} \cdot \frac{4}{3}$	$\frac{8}{21}$
$8\frac{2}{3} \div \frac{5}{6}$	$\frac{26}{3} \div \frac{5}{6}$	$\frac{26}{3} \cdot \frac{6}{5}$	$\frac{156}{15} = \frac{52}{5}$ or $10\frac{2}{5}$
$4\frac{4}{5} \div 1\frac{1}{2}$	$\frac{24}{5} \div \frac{3}{2}$	$\frac{24}{5} \cdot \frac{2}{3}$	$\frac{48}{15} = \frac{16}{5}$ or $3\frac{1}{5}$

2. Find each quotient using your preferred method. Simplify when possible.

a. $\frac{3}{4} \div \frac{9}{14}$

$$\begin{aligned} &\frac{3}{4} \cdot \frac{14}{9} \\ &= \frac{42}{36} \\ &= \frac{7}{6} \text{ or } 1\frac{1}{6} \end{aligned}$$

b. $6\frac{7}{8} \div \frac{5}{7}$

$$\begin{aligned} &\frac{55}{8} \div \frac{5}{7} \\ &= \frac{55}{8} \cdot \frac{7}{5} \\ &= \frac{385}{40} \\ &= \frac{77}{8} \text{ or } 9\frac{5}{8} \end{aligned}$$



Consider highlighting products in this practice that can be simplified, like the bottom two rows of the table in question 1 and both parts of question 2. Direct student attention to fractions whose numerator and denominator share a common factor to help them recognize when products can be simplified.



Consider asking students:

- What do you notice about the numerator and denominator in the product?
- How do you know a fraction can be simplified?
- Why do we simplify fractions and mixed numbers?

1.7.4 Your Turn!

Component Overview: Students practice finding quotients and simplifying when possible. Consider having students work in partners and using station rotations or having a “scavenger hunt” to search for and solve each problem to switch up procedural practice. If using station rotations or the “scavenger hunt,” be sure to write each problem below on a piece of paper and place in various locations around the classroom before beginning.



1.7.4 Your Turn!

1. Find each quotient. Simplify when possible.

a. $\frac{1}{18} \div \frac{1}{9}$

$$\begin{aligned} &\frac{1}{18} \cdot \frac{9}{1} \\ &= \frac{9}{18} \\ &= \frac{1}{2} \end{aligned}$$

b. $2\frac{2}{5} \div \frac{1}{10}$

$$\begin{aligned} &\frac{12}{5} \div \frac{1}{10} \\ &= \frac{12}{5} \cdot \frac{10}{1} \\ &= \frac{120}{5} \\ &= 24 \end{aligned}$$

c. $\frac{3}{15} \div 6$

$$\begin{aligned} &\frac{3}{15} \div \frac{6}{1} \\ &= \frac{3}{15} \cdot \frac{1}{6} \\ &= \frac{3}{90} \\ &= \frac{1}{30} \end{aligned}$$

d. $9 \div \frac{3}{2}$

$$\begin{aligned} &\frac{9}{1} \div \frac{3}{2} \\ &= \frac{9}{1} \cdot \frac{2}{3} \\ &= \frac{18}{3} \\ &= 6 \end{aligned}$$

e. $1\frac{5}{9} \div 5\frac{5}{8}$

$$\begin{aligned} &\frac{14}{9} \div \frac{45}{8} \\ &= \frac{14}{9} \cdot \frac{8}{45} \\ &= \frac{112}{405} \end{aligned}$$

f. $\frac{1}{20} \div \frac{1}{4}$

$$\begin{aligned} &\frac{1}{20} \cdot \frac{4}{1} \\ &= \frac{4}{20} \\ &= \frac{1}{5} \end{aligned}$$

g. $\frac{4}{12} \div \frac{1}{7}$

$$\begin{aligned} &\frac{4}{12} \cdot \frac{7}{1} \\ &= \frac{28}{12} \\ &= \frac{14}{6} \text{ or } 2\frac{1}{3} \end{aligned}$$

h. $2 \div 3\frac{4}{15}$

$$\begin{aligned} &\frac{2}{1} \div \frac{49}{15} \\ &= \frac{2}{1} \cdot \frac{15}{49} \\ &= \frac{30}{49} \end{aligned}$$

i. $1\frac{6}{10} \div 5\frac{3}{4}$

$$\begin{aligned} &\frac{16}{10} \div \frac{23}{4} \\ &= \frac{16}{10} \cdot \frac{4}{23} \\ &= \frac{64}{230} \\ &= \frac{32}{115} \end{aligned}$$



Consider a variety of methods for implementation.

- **Scavenger or Stations:** Station rotation or “scavenger hunt” activity with a card for each problem hidden around the room for students to find and solve.
- **Student Choice:** Students opting in to self-select quotients to find, increasing student choice and providing important observational feedback for which problems students choose and why.
- **Task Cards:** Print and sort problems into Task Cards using the categories of improper, mixed, or other types of fractions and have students practice by category.
- **Trashketball:** Student favorite – each correct response earns a chance to shoot a ball of paper into the can. Most made baskets means a homework pass!

1.7.5 Wrap-Up: Ticket Out the Door

Component Overview: Consider using all or parts of this exercise as a formative assessment to gather information on student learning from this lesson and the last lesson. This Wrap-Up assesses students' ability to both multiply and divide fractions using any method and simplifying when possible. Students should not use a calculator to complete the Wrap-Up.



1.7.5 Wrap-Up: Ticket Out the Door

1. Find each product or quotient. Simplify when possible.

a. $\frac{2}{11} \div \frac{1}{3}$

$$\frac{2}{11} \cdot \frac{3}{1} = \frac{6}{11}$$

b. $\left(\frac{2}{23}\right)\left(3\frac{3}{23}\right)$

$$\frac{2}{23} \cdot \frac{72}{23} = \frac{144}{529}$$



Consider adding complexity by having students solve one or more of the problems in two ways: with the standard algorithm and with a diagram or model. For dividing fractions, students can use a tape diagram as modeled in this lesson. For multiplying fractions, students can use an area model.

c. $2\frac{27}{32} \times \frac{3}{182}$

$$\frac{91}{32} \cdot \frac{3}{182} = \frac{3}{64}$$

d. $3\frac{3}{4} \div 4\frac{1}{6}$

$$\begin{aligned} \frac{15}{4} \div \frac{25}{6} &= \frac{15}{4} \cdot \frac{6}{25} \\ &= \frac{90}{100} \\ &= \frac{9}{10} \end{aligned}$$